# The International Joint Conference on Neural Networks

## 2015 Conference Program

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# International Joint Conference on Neural Networks (IJCNN) 2015

# Program

## Killarney, Ireland (Cill Airne, Éire)
## July 12 – July 17, 2015

Organized and Sponsored by INNS and IEEE-CIS

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# 1 Welcome Messages

## 1.1 Welcome Message from the General Chair and Program Chair of IJCNN 2015

De-Shuang Huang
General Chair

Yoonsuck Choe
Program Chair

On behalf of the Program Committee and Organizing Committee, we would like to warmly welcome you to the 2015 International Joint Conference on Neural Networks (IJCNN 2015) in beautiful Killarney, Ireland.

This conference continues the tradition of joint sponsorship of IJCNN by the International Neural Network Society (INNS) and the IEEE Computational Intelligence Society (IEEE-CIS). We would like to thank the leadership of two organizations, and particularly the Presidents, Ali Minai and Xin Yao, for making this possible.

For IJCNN 2015 we received 834 submissions from 66 countries, 32 of which were later withdrawn. Of these, 550 papers (65.9%) were accepted. The conference features 319 oral presentations, 231 poster presentations, and 9 plenary or invited talks. The program also features 20 tutorials, 5 post-conference workshops, 16 special sessions, and 4 competitions.

The plenary talks by Lee Giles, Marios Polycarpou, Vincenzo Piuri, Barak Pearlmutter, Steve Furber, and Giacomo Rizzolatti as well as three invited speeches by Vladimir Cherkassky, Anders Sandberg, Cesare Alippi reflect the broad themes of neuroscience and engineering applications, and go beyond traditional neural networks into areas like behavioral economics, bio-molecules, robotics, and control, etc. This year's conference has an especially strong set of presentations in the areas of big data and deep learning.

To organize a conference such as this was very demanding in terms of organizational effort, coordination and positive spirit. We would like to thank most of all the members of the Conference Executive Committee, the two Technical Program Co-Chairs, Haibo He and Asim Roy whose dedication made the organization of the conference possible. Special thanks go to Publicity Co-Chairs, Bill Howell, Yun Raymond Fu and Giacomo Boracchi, Tutorials Chair, Martin McGinnity, Publication Chair, Amir Hussain, Plenary Chair, Carlo Francesco Morabito, Web Reviews Chair, Tomasz Cholewo, and Webmaster, Jaerock Kwon. We would also like to thank our Special Session Co-Chairs, Mike Gashler and Jose Garcia-Rodriguez, Competitions Chair, Abir Hussain, Workshops Chair, Pierre-Yves Oudeyer, Awards Chair, Plamen Angelov, Panels Chair, Juyang Weng, Local Arrangements Co-Chairs, Marion McAfee and Kang Li, and Registration Chair, Khan Iftekharuddin. We also thank all other members of the Organizing Committee for their great help, all of whom put in a special effort towards the meeting's success.

We are particularly grateful to all members of the Program Committee and all reviewers who deserve great appreciation for providing discerning and timely reviews of over 830 submissions. In particular, we would like to thank all the authors for contributing their papers. Without the high-quality submissions from the authors, the success of the conference would not have been possible. We are also very grateful for the support we have received from many members of the INNS Board of Governors, and especially for sage advice from Ali Minai, Danil Prokhorov, and Dave Casasent of INNS leadership. Finally, we would also like to thank the organizational team at INNS Central Office led by Marianne Van Wagner and Alison Watson, without whose work the meeting would have been infinitely harder to organize.

As in past years, INNS and IEEE-CIS have provided support for many students to attend the conference. We hope that IJCNN will continue to serve as a place where young researchers can find both knowledge and inspiration.

Finally, we would like to thank the following sponsors for their generous support: Gold sponsors Fáilte Ireland National Tourism Development Authority (www.failtreireland.ie) and Meet in Ireland (meetinireland.com); and Bronze sponsors Budapest Semester in Cognitive Science, Brain-Mind Institute, and Springer.

We wish you a wonderful, productive, and pleasant IJCNN 2015.

| | |
|---|---|
| De-Shuang Huang | Yoonsuck Choe |
| Tongji University, China | Texas A&M University, USA |
| General Chair, IJCNN 2015 | Program Chair, IJCNN 2015 |

## 1.2 Welcome Message from the President of INNS

Ali Minai
INNS President

As President of the International Neural Network Society (INNS), it is a great pleasure and an honor for me to welcome you to the 2015 International Joint Conference on Neural Networks (IJCNN 2015) in beautiful Killarney, Ireland. This conference epitomizes the interdisciplinary nature of the field of neural networks, featuring presentations across a broad set of areas, including neuroengineering, distributed intelligence, computational neuroscience, cognitive modeling, bioinformatics, robotics, and many others. It also features a larger than usual set of plenary and invited talks by distinguished researchers who are shaping these fields. Another important feature of the conference is that it provides a forum where young, aspiring researchers from around the world can meet each other and interact with the leading practitioners in their fields, possibly establishing fruitful new collaborations for the future. All these aspects reflect the International Neural Network Society's fundamental commitment to fostering interdisciplinary research and scientific collaboration.

This conference also continues a successful tradition of collaboration between the INNS and the IEEE Computational Intelligence Society (IEEE-CIS) in their joint organization of IJCNN. Over a period of more than a quarter century, this collaboration has made IJCNN the premiere international conference for researchers in neural networks and related fields. We look forward to continuing this collaboration in Vancouver next year, and for years thereafter.

Over the last year, INNS has made significant strides in outreach to members, co-sponsoring conferences across the world, and promoting the activities of special interest groups. In perhaps its most ambitious new initiative in many years, INNS is organizing the first INNS Conference on Big Data in San Francisco this year (August 8–10, 2015), which will feature outstanding speakers and workshops in this critical area. The Society is also planning to pursue other ways to bring greater value to its members. An important part of this effort is a planned initiative to promote greater diversity within the Society and within the neural networks community in general.

Every year, INNS presents awards to researchers who have distinguished themselves through their contributions to the field of neural computation and its applications. This year, I am proud to announce the following recipients:

- **Hebb Award** recognizing achievement in biological learning: **Michael Hasselmo**
- **Gabor Award** recognizing achievement in engineering applications: **Donald Wunsch**
- **INNS Young Investigator Award**: **Huanhuan Chen**

At a time when a deeper understanding of the brain is being recognized across the world as one of the major "grand challenge" problems for science, researchers in neural networks and related fields represent an indispensable resource. From our position at the intersection of biology and technology, our efforts in understanding the neural basis of cognition, explaining the processes underlying mental illness, elucidating the mechanics of human creativity, helping to build smart prosthetics for the disabled, and creating truly intelligent machines hold the promise of changing the world for the better. The science and technology of the brain has never been more important, or more productive, as the presentations at this conference show.

I want to acknowledge the immense effort put in by the IJCNN'2015 Organizing Committee, led by General Chair De-Shuang Huang and Program Chair Yoonsuck Choe – ably helped by the Technical Co-Chairs, Asim Roy and Haibo He, and all the other chairs and liaisons. Their efforts have made this an exciting conference, with the largest number of attendees of any previous stand-alone IJCNN meeting. And finally, I would like to thank all the participants of IJCNN'2015 for contributing to this great meeting in many different ways. Welcome to Killarney!

Ali Minai
President of INNS

## 1.3 Welcome Message from the President of IEEE-CIS

Xin Yao
IEEE-CIS President

As the President of the IEEE Computational Intelligence Society (CIS), I am delighted to welcome you all to IJCNN'15 in Killarney, Ireland! CIS has been a partner with INNS in running IJCNN each year for more than two decades. The two organisations have collaborated harmoniously and worked closely together to bring IJCNN to the international communities. INNS is leading the organisation of IJCNN'15 and the next year's IJCNN'16 will be led by IEEE CIS as part of the WCCI'2016 in Vancouver, Canada (www.wcci2016.org).

Neural networks have come a long way as a thriving research field. In late 1980s, more papers presented at the neural network conferences were related to training algorithms and some applications in pattern classification. Now the field has grown significantly in both depth and breadth. It has served as a breeding ground for many fast-developing areas, including deep learning and big data analytics. The field has fused with many other fields, such as fuzzy systems, evolutionary computation, computational neuroscience, etc., to form a much broader field of computational intelligence. The old IEEE Transactions on Neural Networks was renamed to IEEE Transactions on Neural Networks and Learning Systems (TNNLS) a few years ago to reflect the development of the field of neural networks and the growing emphasis on learning systems.

Glancing through the programme of IJCNN'15, one can find a very interesting range of topics related to neural networks, from statistical machine learning to computational neuroscience, and from intelligent transport systems to smart industry. The IJCNN'15 programme is a real feast to anyone's brain. I hope every participant will find the conference as exciting as I feel. I actually have a hard time in planning which sessions I should attend, because there are so many interesting sessions (including tutorials) that I want to attend but cannot due to time clashes. I wish I could develop an artificial neural network system, which would attend the session I miss for me and then summarise the key ideas and results back to me after the session.

I recently wrote my President's Message in the May issue of IEEE Computational Intelligence Magazine, questioning the scientific base of ranking conferences in some countries. For me personally, attending a conference is far more than just publishing a paper. It is the whole experience of attending the conference, listening to people describing their work, debating ideas, and interacting with colleagues, that make a conference a conference, not just an outlet for publications. Don't we all have the experience that a new research interest or idea was sparked by some casual conversations at a conference? How could we rank that scientifically?

Wish you all have a wonderful conference experience at IJCNN'15!

Xin Yao
President
IEEE Computational Intelligence Society

# 2 Organizing Committee

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**Program Chair**
Yoonsuck Choe, *Texas A&M University, USA*
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Asim Roy, *Arizona State University, USA*

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**Competition Chair**
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# 4 Reviewers

Note: (1) Organizing committee and program committee members who reviewed papers are also listed. (2) Author last names appear as entered on the submission form.

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Keck, Ingo R.
Kerdvibulvech, Chutisant
Khalifa, Yaser
Khan, Salman
Khorasani, Kash
Khosla, Rajiv
Kilby, Jeff
Kil, Rhee Man
Kim, Jisung
Kim, Jong Kyoung
Kim, Kyung-Joong
Kim, Saehoon
Kim, Seong-Joo
Kinto, Eduardo Akira
Kiselev, Mikhail
Kitani, Edson
Kochhar, Dev S.
Koene, Randal
Kollias, Stefanos
Komendantskaya, Ekaterina
Kong, Adams Wai-Kin
Koo, Imhoi
Koprinkova-Hristova, Petia
Kordik, Pavel
Ko, Sakai
Kosmopoulos, Dimitrios
Kotropoulos, Constantine
Kouroupetroglou, Georgios
Koutroumbas, Konstantinos
Koychev, Ivan
Kramer, Kathleen
Kreinovich, Vladik
Krempl, Georg
Kristensen, Terje
Krohling, Renato
Krzyzak, Adam
Kubota, Naoyuki
Kuh, Anthony
Kulkarni, Anand
Kumarappan, N.
Kumar, Pradeep
Kuremoto, Takashi
Kurfess, Franz
Kurita, Takio
Kuroe, Yasuaki
Kursun, Olcay
Kwon, Jaerock
Laha, Arijit
Lai, Daniel
Lambert-Torres, Germano
Lamb, Luis
Lamirel, Jean-Charles
Langone, Rocco
Lan, Leu-Shing
Lan, Man
Lansner, Anders
Lazaro, Marcelino
Lazhar, Labiod
Lebbah, Mustapha
Lee, Chien-Cheng
Lee, Hyekyoung
Lee, John
Lee, Jong-Seok
Lee, Minho
Lee, Minwoo
Lee, Vincent Cheng-Siong
Leighty, Brian
Lemaire, Vincent
Leray, Philippe
Levine, Daniel
Levy, Pierre
Liang, Hualou
Liang, Zhao
Li, Baichuan
Li, Cheng
Li, Chengdong
Li, Cong
Lightbody, Gordon
Li, Hongliang
Li, Jianmin
Li, Kang
Likas, Aristidis
Lima, Clodoaldo
Lima, Tiago
Lim, Heejin Lim
Linares-Barranco, Alejandro
Lin, Daw-Tung
Linh, Tran Hoai
Li, Ning
Lin, Wei-Song
Liparulo, Luca
Li, Peng
Lippi, Marco
Lisboa, Paulo
Liu, Derong
Liu, Qingfang
Liu, Xiwei
Liu, Yang
Liu, Zhenwei
Livi, Lorenzo
Li, Yuhua
Li, Zhen
Lo Bosco, Giosue'
Loconsole, Claudio
Loefstroem, Tuve
Lo, James T.
Loo, ChuKiong
Looney, David
Lopes, Noel
Lopez de Luise, Daniela
Lopez Lazaro, Jorge
Lourenco, Carlos
Lowrie, Christopher
Loyola R., Diego G.
Lucas, Tarcisio
Luciw, Matthew
Ludermir, Teresa
Luengo, David
Luo, Xiao
Luo, Xiong
Lu, Qiang
Lv, Jiancheng
Lv, Le
Macau, Elbert
Maccio', Danilo

Madeiro, Francisco
Magdalena, Luis
Mahmud, Mufti
Malliaris, Mary
Mandziuk, Jacek
Mao, Hongwei
Marocco, Davide
Martin-Guerrero, Jose D.
Ma, Tiedong
Matsui, Nobuyuki
Matthews, Stephen
Meer, Marius van der
Melin, Patricia
Mendoza, Leonardo
Miikkulainen, Risto
Morell, Vicente
Muller, Daniel Nehme
Mylonas, Phivos
Nakano, Ryohei
Narayan, Vikram
Nichols, Barry
Nishimura, Haruhiko
Ntalampiras, Stavros
Oh, Sang-Hoon
Okun, Oleg
Olteanu, Madalina
Ortega, Juan A.
Osinenko, Pavel
Ozawa, Seiichi
Paetz, Juergen
Palmieri, Francesco
Pan, Shing-Tai
Pappalardo, Francesco
Passow, Benjamin
Pavlidou, Meropi
Pena, Marian
Peralta Donate, Juan
Peres, Sarajane
Peterson, Leif E.
Petrovskiy, Mikhail
Piastra, Marco
Pikrakis, Aggelos
Piuri, Vincenzo
Porrmann, Mario
Prati, Ronaldo
Proietti, Andrea
Psarrou, Alexandra
Qin, Bin
Qiu, Fang
Radtke, Paulo Vinicius Wolski
Ramadan, Rabie
Raugi, Marco
Redko, Ievghen
Rezaei, Siamak
Rivas-Perea, Pablo
Rizzo, Riccardo
Rojo-Alvarez, Jose Luis
Rosa, Joao Luis G.
Rotstein, Horacio
Rudolph, George
Madevska Bogdanova, Ana
Maggioni, Matteo
Mali, Kalyani
Mallipeddi, Rammohan
Maniadakis, Michail
Marchese, Luca
Martin, Arnaud
Masafumi, Hagiwara
Matsubara, Edson Takashi
Matsui, Tetsuo
Medeiros, Talles Henrique de
Mehrkanoon, Siamak
Mello, Carlos
Mentzelopoulos, Markos
Minku, Leandro
Mori, Hiroki
Murphey, Yi
Naidoo, Bashan
Narayanan, Arun
Neme, Antonio
Nikolik, PhD
Nishio, Yoshifumi
Oguz, Cihan
Okada, Shogo
Oliveira, Luiz
Oneto, Luca
Orts-Escolano, Sergio
Osoba, Osonde
Paasio, Ari
Palensky, Peter
Panagiotopoulos, Dimokritos
Papakostas, George
Parsapoor, Mahboobeh
Patane', Luca
Pears, Russel
Peng, Hanxiang
Pereira, Carlos
Perrone, Antonio Luigi
Petia, Georgieva
Pham, Cong-Kha
Piazza, Francesco
Pilly, Praveen
Pokrajac, David
Powell, Warren
Precup, Radu-Emil
Prokhorov, Danil
Purnomo, Mauridhi Hery
Qin, Chunbin
Qi, Zhou
Rai, Shri
Rani, Sobha
Raza, Haider
Regazzoni, Francesco
Ridella, Sandro
Rivas Santos, Victor Manuel
Rodriguez Rivero, Cristian
Romero, Roseli
Rossello, Josep L.
Roux, Ludovic
Rueckert, Ulrich
Madureira, Ana Maria
Maguire, Liam
Malik, Om
Mandic, Danilo
Mann, Timothy
Marchi, Erik
Martinelli, Eugenio
Masulli, Francesco
Matsuda, Satoshi
Matsuyama, Yasuo
Medsker, Larry
Meindl, Tassilo
Melo-Pinto, Pedro
Mi, Guyue
Mohagheghi, Salman
Muhamad Amin, Anang Hudaya
Murtagh, Fionn
Nakada, Yohei
Narayan, Sridhar
Nguyen, Minh Nhut
Ninomiya, Hiroshi
Ni, Zhen
Ohkura, Kazuhiro
O'Keefe, Simon
Olsen, Megan
Ong, Yew-Soon
OShea, James
Oswald, Cyril
Pachidis, Theodore
Palm, Guenther
Panella, Massimo
Paplinski, Andrew P
Pasa, Leandro
Patan, Krzysztof
Pedrycz, Witold
Peng, Xi
Pereira, Ivo
Peter, Andras
Petridis, Stavros
Phlypo, Ronald
Picton, Phil
Pino, Robinson
Polikar, Robi
Pozo, Aurora
Principi, Emanuele
Prudencio, Ricardo
Qian, Jianjun
Qin, Sitian
Quiles, Marcos
Rajati, Mohammad Reza
Rast, Alexander
Recamonde Mendoza, Mariana
Ren, Zijian
Ripon, Kazi Shah Nawaz
Rizzi, Antonello
Rogovschi, Nicoleta
Romoli, Laura
Roth, Peter
Roveri, Manuel
Ruiz Llata, Marta

Ryu, Keun Ho
Sahba, Farhang
Saito, Toshimichi
Sakurai, Shigeaki
Salleb-Aouissi, Ansaf
Sanguineti, Marcello
Santos, Sergio P.
Sassi, Roberto
Saval, Marcelo
S. Cardoso, Jaime
Scherer, Rafal
Schuller, Bjoern
Seixas, Jose Manoel De
Serpa, Alexandre Luiz de Oliveira
Sewell, Martin Victor
Shen, Furao
Shen, Yun
Shieh, Chin-Shiuh
Shin, Jungpiil
Sigelle, Marc
Sillitti, Alberto
Silva, Paulo
Sitte, Joaquin
Smith, Leslie
Sona, Diego
Song, Ruizhuo
Souza, Jackson
Squartini, Stefano
Srinivasa, Narayan
Staiano, Antonino
Sublime, Jeremie
Sun, Bing-Yu
Sun, Liang
Surampudi, Durga Bhavani
Suykens, Johan
Taha, Tarek
Takahashi, Takashi
Tan, Boon Hwa
Tang, Yufei
Tattoli, Giacomo
Teschl, Reinhard
Thumati, Balaje
Tiumentsev, Yury
Torres-Huitzil, Cesar
Trovo', Francesco
Tsuruta, Nayouki
Turchenko, Volodymyr
Uddin, Muhammad
Valdez, Fevrier
Valverde-Albacete, Francisco J.
Vasilakos, Athanasios
Vega, Javier
Vellasco, Marley
Veltri, Pierangelo
Verma, Brijesh
Viktor, Herna
Vitay, Julien
Vrabie, Draguna
Wall, Julie
Wang, Ding
Sabourin, Robert
Saighi, Sylvain
Sakr, Nizar
Salatino, Angelo Antonio
Sanchez Giraldo, Luis Gonzalo
Sansone, Carlo
Sarangapani, Jagannathan
Sato, Shigeo
Sazonov, Edward
Scarpiniti, Michele
Schleif, Frank
Schumann, Johann
Sergio, Anderson
Sethi, Ishwar
Shannon, Thaddeus
Shen, Yi
Shibata, Katsunari
Shim, Eunsoo
Shouno, Hayaru
Si, Jennie
Silva, Catarina
Silva, Thiago Christiano
Skabar, Andrew
Solari, Fabio
Song, Chao
Soschen, Alona
Sperduti, Alessandro
Sramka, Michal
Srinivasan, Soundararajan
Ster, Branko
Subramanian, Kartick
Sundararajan, Narasimhan
Sun, Tsung-Ying
Suresh, Sundaram
Swarup, K. Shanti
Taheri, Javid
Tambouratzis, Tatiana
Tang, Bin
Tan, Mingkui
Teegavarapu, Ramesh
Thivierge, Jean-Philippe
Timoszczuk, Antonio Pedro
Tivive, Fok Hing Chi
Torres-Sospedra, Joaquin
Tsang, Ing Ren
Tuckova, Jana
Tu, Zhengwen
Uosaki, Katsuji
Valdivia, Karina
Vamvoudakis, Kyriakos G.
Vassiljeva, Kristina
Velastin, Sergio
Vellasco, Pedro
Ventresca, Mario
Verri, Filipe Alves Neto
Vincent, Nicole
Volosencu, Constantin
Vrana, Stanislav
Wan, Feng
Wang, Hung-Jen
Sadeghian, Alireza
Saito, Jose Hiroki
Sakurai, Akito
Salazar, Addisson
Sandmann, Humberto
Santana Junior, Orivaldo
Sarmiento Vega, Auxiliadora
Satyadas, Antony
Scardapane, Simone
Schaefer, Gerald
Schleif, Frank-Michael
Schwenker, Friedhelm
Seridi, Hamid
Severini, Marco
Shell, Jethro
Shen, Yuan
Shi, Bertram
Shim, Vui Ann
Siano, Pierluigi
Sil, Jaya
Silva, Leandro Augusto
Singh, Abhishek
Slot, Krzysztof
Solteiro Pires, Eduardo
Song, Insu
Sousa, Fabiano
Spratling, Michael
Sreela, Sasi
Stafylopatis, Andreas
Stroppa, Fabio
Sulistijono, Indra Adji
Sung, Chul
Sun, Zhanqquan
Suri, Manan
Szu, Harold
Takahashi, Norikazu
Tanaka, Toshihisa
Tang, Bo
Taskaya-Temizel, Tugba
Terekhov, Sergey
Thomaz, Carlos
Tinos, Renato
Tizhoosh, Hamid R.
Troncoso, Alicia
Tsoy, Yury
Tu, Kun
Twala, Bhekisipho
Urio, Paulo Roberto
Valerio, Lorenzo
Vasconcelos, Germano
Vasudevan, Bintu
Velde, Frank van der
Vellido, Alfredo
Verdes, Pablo F.
Vigneron, Vincent
Virtanen, Tuomas
Von Zuben, Fernando J.
Wahde, Mattias
Wang, DeLiang
Wang, Jianyong

Wang, Lei
Wang, Lipo
Wang, Pan
Wang, Qian
Wang, Sheng
Wang, Wenwu
Wang, Xiaoping
Wang, Xinying
Wang, Yijun
Wang, Yu-Xiong
Wang, Yuxuan
Wang, Zhanshan
Wang, Zhiliang
Watts, Michael
Wei, Wei
Weng, John
Wermter, Stefan
Wichern, Gordon
Wiering, Marco
Wong, Kok Wai
Wong, Man Leung
Wozniak, Michal
Wu, Chenxia
Wuertz, Rolf
Wu, Lei
Wunsch, Donald
Wurtz, Rolf
Wu, Yunfeng
Xiang, Tao
Xiao, Geyang
Xiao, Qinghan
Xia, Yousheng
Xia, Yuanqing
Xie, Songyun
Xin, Xin
Xu, Dongming
Xue, Xiangyang
Xu, Jin
Xu, Jun
Xu, Xin
Yamakawa, Takeshi
Yang, Peipei
Yang, Shaofu
Yang, Zhile
Yan, Jun
Yeh, Wei-Chang
Yi, Xinlei
Yoo, Jaewook
Young, Steven
Yu, Qiang
Zanchettin, Cleber
Zegers, Pablo
Zeleznikow, John
Zeng, Zhigang
Zhang, Botao
Zhang, Dehua
Zhang, Huaguang
Zhang, Long
Zhang, Ruiliang
Zhang, Xiaohui
Zhang, Yanming
Zhang, Yi
Zhang, Zhao
Zhao, Bo
Zhao, Dongbin
Zhao, Liu
Zhao, Qiangfu
Zhao, Xiaodong
Zheng, Ren
Zhou, Huiyu
Zhou, Shang-Ming
Zhou, Tom Chao
Zhu, Pingping
Zou, Weibao
Zunino, Rodolfo
Zuo, Xingquan

# 5 INNS Organization

## 5.1 2014 INNS Officers (Executive Committee)

| | |
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| **President** | |
| Ali Minai | University of Cincinnati |
| **Vice-President for Membership** | |
| Irwin King | The Chinese University of Hong Kong |
| **Vice-President for Conferences** | |
| Danil Prokhorov | TRINA, Toyota Technical Center |
| **Treasurer** | |
| David Casasent | Carnegie Mellon University |
| **Secretary (-6/2015)** | |
| William Howell | Natural Resources Canada (retired) |
| **Secretary (7/2015-12/2016)** | |
| Yoonsuck Choe | Texas A&M University |

## 5.2 2015 Board of Governors

| | |
|---|---|
| Cesare Alippi | Politecnico di Milano |
| Plamen Angelov | Lancaster University |
| Richard Duro | EPS, Universidad Coruna |
| Peter Erdi | Kalamazoo College; Hungarian Academy of Sciences |
| William Howell | |
| De-Shuang Huang | Tongji University |
| Nikola Kasabov | Knowledge Engineering and Discovery Research Institute |
| Irwin King | The Chinese University of Hong Kong |
| Teresa Ludermir | Universidade Federal de Pernambuco |
| Danilo Mandic | Imperial College |
| Risto Miikkulainen | University of Texas at Austin |
| Ali Minai | University of Cincinnati |
| Leonid Perlovsky | AFRL/SN |
| Asim Roy | Arizona State University |
| Juergen Schmidhuber | The Swiss AI Lab IDSIA |
| Hava Siegelmann | University of Massachusetts |
| Ron Sun | Rensselaer Polytechnic Institute |
| Marley Vellasco | Rua Marques De Sao Vicente |
| DeLiang Wang | Ohio State University |
| Lipo Wang | Nanyang Technological University |

## 5.3 INNS Committees

| | |
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| **Nomination Committee Chair** | |
| Ron Sun | Rensselaer Polytechnic Institute |
| **Award Committee Chair** | |
| Juergen Schmidhuber | The Swiss AI Lab IDSIA |
| **Government and Corporate Liaison Committee Chair** | |
| Ali Minai | University of Cincinnati |
| **Publication Committee Chair** | |
| Soo-Young Lee | Korea Advanced Institute of Science and Technology (KAIST) |
| **Newsletter/Magazine Editor** | |
| Soo-Young Lee | Korea Advanced Institute of Science and Technology (KAIST) |
| **Education Activities Committee Chair** | |
| Timo Honkela | University of Helsinki |
| Juyang (John) Weng | Michigan State University |

# 6 IEEE CIS Organization

## 6.1 Executive committee

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| **President** | |
| Xin Yao | University of Birmingham |
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| Yaochu Jin | University of Surrey |
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| **Vice President for Member Activities** | |
| Pau-Choo (Julia) Chung | National Cheng Kung University |
| **Vice President for Education** | |
| Cesare Alippi | Politecnico di Milano |

# 7 Featured Talks: Plenary and Invited Talks

## 7.1 Plenary Speakers

- Giacomo Rizzolatti, *University of Parma, Italy*
  - The Double Life of the Motor System: Action Production and Action Understanding
  - Monday July 13, 2015, 8am-9am
- Marios Polycarpou, *University of Cyprus, Cyprus*
  - Fault Detection and Isolation in Uncertain Big-Data Environments
  - Tuesday July 14, 2015, 8am-9am
- Vincenzo Piuri, *University of Milan, Italy*
  - Computational Intelligence Technologies for 3D Surface Reconstruction
  - Tuesday July 14, 2015, 1:30pm-2:30pm
- Barak Pearlmutter, *National University of Ireland, Ireland*
  - Critical Dynamics and Pathological Phenomena in the Brain
  - Wednesday July 15, 2015, 8am-9am
- Stephen Furber, *University of Manchester, United Kingdom*
  - The SpiNNaker Project
  - Wednesday July 15, 2015, 1:10pmam-2:10pm
- Lee Giles, *Pennsylvania State University, United States*
  - Machine Learning and Data Mining for Scholarly Big Data
  - Thursday July 16, 2015, 8am-9am

## 7.2 Invited Speakers

- Anders Sandberg, *Oxford University*
  - Ethics and large-scale neural simulations: when do we need to start caring for networks, rather than about them?
  - Monday July 13, 2015, 10:50am-11:30am
- Vladimir Cherkassky, *University of Minnesota*
  - Methodological Aspects of VC-theory
  - Tuesday July 14, 2015, 10:50am-11:30am
- Cesare Alippi, *Politecnico di Milano, Italy*
  - Intelligence for Cyber-Physical and Embedded Systems
  - Wednesday July 15, 2015, 10:50am-11:30am

# 8 Program Overview

**Sunday, July 12th, 2015**

| Time | **Brehon 1:** | **Brehon 2 :** | **Ross :** | **Park Suite:** | **Torc :** |
|---|---|---|---|---|---|
| 8:00AM | **tut1:** Tutorial 1: Forecasting with Recurrent Neural Networks: 12 Tricks | **tut2:** Tutorial 2: The Mind-Brain, Big Data and Automomous Learning | **tut3:** Tutorial 3: Noninvasive Electroencephalogram -based Brain-Computer Interfaces | **tut4a:** Tutorial 4(a): Robust Model-based Learning: Methods, Algorithms and Applications | **tut4b:** Tutorial 4(b):Simulating an entire nervous system? An exemplary Caenorhabditis elegans emulation case study |
| 10:00AM | Break | | | | |
| 10:30AM | **tut5:** Tutorial 5: Computational Intelligence for Wearable Physiological Sensing | **tut6:** Tutorial 6: Compositionality and Self-Organization in Cognitive Minds: Lessons from Neuro-Robotics Experimental Studies | **tut7:** Tutorial 7: Computational Neuroscience: Past - Present - Future | **tut8:** Tutorial 8: Data visualization with dimensionality reduction and manifold learning | **tut9:** Tutorial 9: Dynamic Systems and Learning in the Model Space |
| 12:30PM | Break | | | | |
| 1:30PM | **tut10:** Tutorial 10: Canceled | **tut11:** Tutorial 11: Multi-Task Learning Primer | **tut12:** Tutorial 12: Learning in indefinite proximity spaces: Mathematical foundations, representations, and models | **tut13:** Tutorial 13: Successful Applications of Neural Networks for Information Fusion | **tut14:** Tutorial 14: Conformal Prediction: A Valid Approach to Confidence Predictions |
| 3:30PM | Break | | | | |
| 4:00PM | **tut15:** Tutorial 15: Advances in Universum Learning | **tut16:** Tutorial 16: Learning Autonomously from Big Data Streams | **tut17:** Tutorial 17: Feature Selection Technique for Gene Expression Data Analysis | **tut18:** Tutorial 18: Spiking Neural Networks in Silicon: From Building Blocks to Architectures of Neuromorphic Systems | **tut19:** Tutorial 19: Learning architectures and training algorithms - comparative studies |
| 6:00PM | Break | | | | |
| 6:30pm | **Opening Reception:** Mangerton Suite, Gleneagle | | | | |
| 7:30PM | End of Day | | | | |

## Monday, July 13th, 2015

| Time | Auditorium: | Ballroom: | Brehon: | Park Suite: | Mangerton: |
|---|---|---|---|---|---|
| 8:00AM | **Plenary1** : Plenary session: Giacomo Rizzolatti (Auditorium) | | | | |
| 9:00AM | Break | | | | |
| 9:10AM | **Chaos:** Nonlinear dynamics and chaos | **ss28:** Clustering and Co-clustering | **Graph:** Graphs | **ss11-1:** Emerging trends in Computational Intelligence methods for Biomedicine and Healthcare | **Neurosc1:** Neuroscience 1: behavioral modeling |
| 10:30AM | Break | | | | |
| 10:50AM | **Invited1:** Invited Speaker: Anders Sandberg | **ss23-1:** Complex and memristive networks 1 | **Manif1:** Manifold learning and dimensionality reduction 1 | **ss11-2:** SS: Biomedical applications | **Dyn1:** Neurodynamics 1 |
| 11:30AM | **Emotion:** Emotion and motivation | **ss23-1:** | **Manif1:** | **ss11-2:** | **Dyn1:** |
| 12:10PM | Break | | | | |
| 1:30PM | **Finance1:** Financial and commercial applications | **EEGMEG1:** EEG/MEG analysis and applications 1 | **Manif2:** Manifold learning and dimensionality reduction 2 | **Sampling:** Sampling | **Dyn2:** Neurodynamics 2 |
| 2:30PM | Break | | | | |
| 2:40PM | **Finance2:** Financial and commercial applications | **ss23-2:** Complex and memristive networks 2 | **DeepNN1:** Deep neural networks 1 | **ss13:** Emerging Methodologies for Big Data Integration | **Neurosc2:** Neuroscience 2: neurons, synapses, and circuits |
| 4:20PM | Break | | | | |
| 4:40PM | **Temporal:** Time and temporal processes | **EEGMEG2:** EEG/MEG analysis and applications 2 | **DeepNN2:** Deep neural networks 2 | **ss26:** Optimizing Neural Networks Using Evolutionary Computation and Swarm Intelligence | **Neurosc3:** Neuroscience 3: behavior, memory, and motivation |
| 6:20PM | Break | | | | |
| 7:30pm | **Poster Session:**INEC Ground Floor and 1st Floor Lobby | | | | |
| 9:30PM | End of Day | | | | |

## Tuesday, July 14th, 2015

| Time | Auditorium: | Ballroom: | Brehon: | Park Suite: | Mangerton: |
|---|---|---|---|---|---|
| 8:00AM | **Plenary2** : Plenary session: Mario Polycarpou (Auditorium) | | | | |
| 9:00AM | Break | | | | |
| 9:10AM | **Social:** Social media analysis | **ss32-1:** Autonomous Machine Learning for Cyber-Physical Systems 1 | **ss17:** Autonomous Learning from Big Data | **Theory1:** Neural networks theory 1 | **Predict:** Prediction and forecasting |
| 10:30AM | Break | | | | |
| 10:50AM | **Invited2:** Invited Speaker: Vladimir Cherkassky | **ss32-2:** Autonomous Machine Learning for Cyber-Physical Systems 2 | **ss25:** Computational Intelligence Applications to Environmental Sustainability and Sustainable Development: Theory and Applications | **Theory2:** Neural networks theory 2 | **Ensemble:** Ensemble learning |
| 11:30AM | **SVM1:** Support Vector Machines 1 | **ss32-2:** | **ss25:** | **Theory2:** | **Ensemble:** |
| 12:10PM | Break | | | | |
| 1:30PM | **Plenary3** : Plenary session: Vincenzo Piuri (Auditorium) | | | | |
| 2:30PM | Break | | | | |
| 2:40PM | **SVM2 :** Support Vector Machines 2 | **RNN:** Recurrent neural networks | **ss03:** Cognition and Development | **Theory3:** Neural networks theory 3 | **ss36:** Ensemble Systems and Machine Learning |
| 4:20PM | Break | | | | |
| 4:40PM | **SVM3:** Support Vector Machines 3 | **ss29:** Modeling and Forecasting Financial and Commodity Markets by Neural Networks | **ss27:** Models of Cognitive-Emotional Interactions | **Language:** Natural language processing | **BioApp:** Biomedical applications |
| 6:20PM | Break | | | | |
| 7:30pm | **Poster Session:**INEC Ground Floor and 1st Floor Lobby | | | | |
| 9:30PM | End of Day | | | | |

## Wednesday, July 15th, 2015

| Time | Auditorium: | Ballroom: | Brehon: | Park Suite: | Mangerton: |
|---|---|---|---|---|---|
| 8:00AM | **Plenary4** : Plenary session: Barak Pearlmutter (Auditorium) | | | | |
| 9:00AM | Break | | | | |
| 9:10AM | **Struct:** Structures and hierarchies | **ss33:** Intelligent Vehicle Systems | **Spiking1:** Spiking neural networks 1 | **Robot1:** Robotics 1: Spatial Cognition and Navigation | **HW1:** Hardware 1 |
| 10:30AM | Break | | | | |
| 10:50AM | **Invited3:** Invited Speaker: Cesare Alippi | **ss12-1:** Concept Drift, Domain Adaptation and Learning in Dynamic Environments 1 | **Spiking2:** Spiking neural networks 2 | **Robot2:** Robotics 2: Sensory and motor processing | **HW2:** Hardware 2 |
| 11:30AM | **ML1:** Machine learning 1 | **ss12-1:** | **Spiking2:** | **Robot2:** | **HW2:** |
| 12:10PM | Break | | | | |
| 1:10PM | **Plenary5** : Plenary session: Steve Furber (Auditorium) | | | | |
| 2:10PM | Break | | | | |
| 2:20PM | **ML2:** Machine learning 2 | **ss12-2:** Concept Drift, Domain Adaptation and Learning in Dynamic Environments 2 | **ss01-1:** Computational Intelligence applied to Vision and Robotics (CIVR) 1 | **RL1:** Reinforcement learning 1 | **HW3:** Hardware 3 |
| 4:00PM | Break | | | | |
| 4:20PM | **ML3:** Machine learning 3 [ROOM CHANGE: INNISFALLEN] | **ss12-3:** Concept Drift, Domain Adaptation and Learning in Dynamic Environments 3 | **ss01-2:** Computational Intelligence applied to Vision and Robotics (CIVR) 2 | **RL2:** Reinforcement learning 2 | **Noise:** Anomaly and noise |
| 6:00PM | Break | | | | |
| 7:00PM | **Banquet:** INEC Ground Floor | | | | |
| 9:30PM | End of Day | | | | |

## Thursday, July 16th, 2015

| Time | **Auditorium:** | **Ballroom:** | **Brehon:** | **Park Suite:** | **Mangerton:** |
|---|---|---|---|---|---|
| 8:00AM | **Plenary6** : Plenary session: Lee Giles (Auditorium) | | | | |
| 9:00AM | Break | | | | |
| 9:10AM | **ss16-1:** Computational Intelligence Algorithms for Digital Audio Applications 1 | **ML4:** Machine learning 4 | **Spiking3:** Spiking neural networks 3 | **Cluster:** Clustering | **BioPerc1:** Biologically inspired perception 1 |
| 10:30AM | Break | | | | |
| 10:50AM | **ss16-2/r:** Computational Intelligence Algorithms for Digital Audio Applications 2 + Regular session on Image Analysis | **Datamine:** Data and text mining | **Image:** Image analysis | **Local :** Local learning | **BioPerc2:** Biologically inspired perception 2 |
| 12:10PM | Break | | | | |
| 1:30PM | | **ws1:** Workshop 1: The 2nd International Workshop on Advances in Learning from/with Multiple Learners (ALML 2015) | **ws2:** Workshop 2: The International Workshop on Spatial Representations in Biology and Robots | **ws3:** Workshop 3: Computational Neurology and Psychiatry: Do we need it? | **ws4:** Workshop 4: BMI Workshop on Brain-Mind 1 |
| 4:30PM | End of Day | | | | |

## Friday, July 17th, 2015

| Time | | | | | **Innisfallen:** |
|---|---|---|---|---|---|
| 9:00AM | | | | | **ws5-1:** Workshop 5: The 2nd International Workshop on Computational Energy Management in Smart Grids (CEMiSG 2015) 1 |
| 12:00PM | Break | | | | |
| 1:30PM | | | | | **ws5-2:** Workshop 5: The 2nd International Workshop on Computational Energy Management in Smart Grids (CEMiSG 2015) 2 |
| 4:30PM | End of Day | | | | |

# 9 Program

## IJCNN 2015 Program

### Sunday, July 12, 2015

**Tutorial tut1: Tutorial 1: Forecasting with Recurrent Neural Networks: 12 Tricks**
Sunday, July 12, 8:00AM-10:00AM, Room: Brehon 1, Instructor: Hans-Georg Zimmermann, Christoph Tietz and Ralph Grothmann

**Tutorial tut2: Tutorial 2: The Mind-Brain, Big Data and Automomous Learning**
Sunday, July 12, 8:00AM-10:00AM, Room: Brehon 2, Instructor: Leonid I. Perlovsky

**Tutorial tut3: Tutorial 3: Noninvasive Electroencephalogram-based Brain-Computer Interfaces**
Sunday, July 12, 8:00AM-10:00AM, Room: Ross, Instructor: Joao Luis Rosa

**Tutorial tut4a: Tutorial 4(a): Robust Model-based Learning: Methods, Algorithms and Applications**
Sunday, July 12, 8:00AM-10:00AM, Room: Park Suite, Instructor: Yixin Chen and Xin Dang

**Tutorial tut4b: Tutorial 4(b):Simulating an entire nervous system? An exemplary Caenorhabditis elegans emulation case study**
Sunday, July 12, 8:00AM-10:00AM, Room: Torc, Instructor: Axel Blau, Martin McGinnity, Fearghal Morgan, Andoni Mujika

**Tutorial tut5: Tutorial 5: Computational Intelligence for Wearable Physiological Sensing**
Sunday, July 12, 10:30AM-12:30PM, Room: Brehon 1, Instructor: Danilo Mandic and Valentin Goverdovsky

**Tutorial tut6: Tutorial 6: Compositionality and Self-Organization in Cognitive Minds: Lessons from Neuro-Robotics Experimental Studies**
Sunday, July 12, 10:30AM-12:30PM, Room: Brehon 2, Instructor: Jun Tani

**Tutorial tut7: Tutorial 7: Computational Neuroscience: Past - Present - Future**
Sunday, July 12, 10:30AM-12:30PM, Room: Ross, Instructor: Peter Erdi

**Tutorial tut8: Tutorial 8: Data visualization with dimensionality reduction and manifold learning**
Sunday, July 12, 10:30AM-12:30PM, Room: Park Suite, Instructor: Michel Verleysen and John A. Lee

**Tutorial tut9: Tutorial 9: Dynamic Systems and Learning in the Model Space**
Sunday, July 12, 10:30AM-12:30PM, Room: Torc, Instructor: Huanhuan Chen and Peter Tino

**Tutorial tut10: Tutorial 10: Canceled**
Sunday, July 12, 1:30PM-3:30PM, Room: Brehon 1, Instructor: N/A

**Tutorial tut11: Tutorial 11: Multi-Task Learning Primer**
Sunday, July 12, 1:30PM-3:30PM, Room: Brehon 2, Instructor: Georgios Anagnostopoulos and Cong Li

**Tutorial tut12: Tutorial 12: Learning in indefinite proximity spaces: Mathematical foundations, representations, and models**
Sunday, July 12, 1:30PM-3:30PM, Room: Ross, Instructor: Peter Tino and Frank-Michael Schleif

**Tutorial tut13: Tutorial 13: Successful Applications of Neural Networks for Information Fusion**
Sunday, July 12, 1:30PM-3:30PM, Room: Park Suite, Instructor: Stephen Stubberud and Kathleen Kramer

**Tutorial tut14: Tutorial 14: Conformal Prediction: A Valid Approach to Confidence Predictions**
Sunday, July 12, 1:30PM-3:30PM, Room: Torc, Instructor: Henrik Bostrom, Alex Gammerman,Ulf Johansson, Lars Carlsson, Henrik Linusson

**Tutorial tut15: Tutorial 15: Advances in Universum Learning**
Sunday, July 12, 4:00PM-6:00PM, Room: Brehon 1, Instructor: Vladimir Cherkassky and Sauptik Dhar

**Tutorial tut16: Tutorial 16: Learning Autonomously from Big Data Streams**
Sunday, July 12, 4:00PM-6:00PM, Room: Brehon 2, Instructor: Plamen Angelov and Asim Roy

**Tutorial tut17: Tutorial 17: Feature Selection Technique for Gene Expression Data Analysis**
Sunday, July 12, 4:00PM-6:00PM, Room: Ross, Instructor: B. Chandra

**Tutorial tut18: Tutorial 18: Spiking Neural Networks in Silicon: From Building Blocks to Architectures of Neuromorphic Systems**
Sunday, July 12, 4:00PM-6:00PM, Room: Park Suite, Instructor: Arindam Basu

**Tutorial tut19: Tutorial 19: Learning architectures and training algorithms - comparative studies**
Sunday, July 12, 4:00PM-6:00PM, Room: Torc, Instructor: Bigdan Wilamowski

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## Monday, July 13, 2015

**Plenary Talk Plenary1: Plenary session: Giacomo Rizzolatti**
Monday, July 13, 8:00AM-9:00AM, Room: Auditorium, Chair: Morabito, Francesco Carlo

**Session Chaos: Nonlinear dynamics and chaos**
Monday, July 13, 9:10AM-10:30AM, Room: Auditorium, Chair: Shanahan, Murray

9:10AM Local and Global Criticality within Oscillating Networks of Spiking Neurons [#15411]

Filipe Peliz Pinto Teixeira and Murray Shanahan

9:30AM Reinforcement Learning with Internal-Dynamics-based Exploration Using a Chaotic Neural Network [#15231]

Katsunari Shibata and Yuta Sakashita

9:50AM Chaotic Multidirectional Associative Memory with Adaptive Scaling Factor of Refractoriness [#15529]

Hayashi Nagamasa and Osana Yuko

10:10AM Prediction Interval-based Neural Network Controller for Nonlinear Processes [#15155]

Mohammad Anwar Hosen, Abbas Khosravi, Saeid Nahavandi, Douglas Creighton and Syed Moshfeq Salaken

**Special Session ss28: Clustering and Co-clustering**
Monday, July 13, 9:10AM-10:30AM, Room: Ballroom, Chair: Lemaire, Vincent; Gisbrecht, Andrej; Lamierl, Jean-Charles

9:10AM Discriminative Dimensionality Reduction for Regression Problems using the Fisher Metric [#15656]

Alexander Schulz and Barbara Hammer

9:30AM Automatic Discovery of Metagenomic Structure [#15327]

Markus Lux, Alexander Sczyrba and Barbara Hammer

9:50AM An Initialization Scheme for Supervized K-means [#15404]

Vincent Lemaire, Oumaima Alaoui Ismaili and Antoine Cornuejols

10:10AM A New Approach for Event Detection using k-means Clustering and Neural Networks [#15677]

Muyiwa Olakanmi Oladimeji, Mikdam Turkey, Mohammad Ghavami and Sandra Dudley

**Session Graph: Graphs**
Monday, July 13, 9:10AM-10:30AM, Room: Brehon, Chair: Doboli, Simona

9:10AM Generalized Label Propagation [#15672]

Asher Hensley, Alex Doboli, Rami Mangoubi and Simona Doboli

9:30AM Robust Multi-class Graph Transduction with Higher Order Regularization [#15080]

Celso Sousa and Gustavo Batista

9:50AM An experimental analysis on time series transductive classification on graphs [#15081]

Celso Sousa, Vinicius Souza and Gustavo Batista

10:10AM Two-Dimensional Multi-layer Factor Graphs in Reduced Normal Form [#15212]

Amedeo Buonanno and Francesco A.N. Palmieri

**Special Session ss11-1: Emerging trends in Computational Intelligence methods for Biomedicine and Healthcare**
Monday, July 13, 9:10AM-10:30AM, Room: Park Suite, Chair: Vellido, Alfredo

9:10AM Selecting target concept in one-class classification for handling class imbalance problem [#15552]

Beatriz Perez-Sanchez, Oscar Fontenla-Romero and Noelia Sanchez-Marono

9:30AM Advanced Classification of Alzheimer's Disease and Healthy Subjects Based on EEG Markers [#15276]

Vitoantonio Bevilacqua, Angelo Antonio Salatino, Carlo Di Leo, Giacomo Tattoli, Domenico Buongiorno, Domenico Signorile, Claudio Babiloni, Claudio Del Percio, Antonio Ivano Triggiani and Loreto Gesualdo

9:50AM The Extracellular N-terminal Domain Suffices to Discriminate Class C G Protein-Coupled Receptor Subtypes from n-Grams of their Sequences [#15481]

Caroline Konig, Rene Alquezar, Alfredo Vellido and Jesus Giraldo

10:10AM A supervised CAD to support telemedicine in hematology [#15277]

Vitoantonio Bevilacqua, Domenico Buongiorno, Pierluigi Carlucci, Ferdinando Giglio, Giacomo Tattoli, Attilio Guarini, Giovanni Simone, Nicola Sgherza, Francesco Girardi, Giacoma De Tullio, Carla Minoia, Anna Scattone, Alfredo Zito and Loreto Gesualdo

**Session Neurosc1: Neuroscience 1: behavioral modeling**
Monday, July 13, 9:10AM-10:30AM, Room: Mangerton, Chair: Parker, Alice

9:10AM Learning Human Motion Feedback with Neural Self-Organization [#15610]

German Ignacio Parisi, Florian Von Stosch, Sven Magg and Stefan Wermter

9:30AM On the utility of sparse neural representations in adaptive behaving agents [#15526]

Thusitha Chandrapala, Bertram Shi and Jochen Triesch

9:50AM Emergence of Tool Use in an Articulated Limb Controlled by Evolved Neural Circuits [#15417]

Qinbo Li, Jaewook Yoo and Yoonsuck Choe

10:10AM Neural Circuits for Touch-Induced Locomotion in Caenorhabditis Elegans [#15201]

Sukanya Patil, Kaidi Zhou and Alice Parker

**Panel Session Invited1: Invited Speaker: Anders Sandberg**
Monday, July 13, 10:50AM-11:30AM, Room: Auditorium, Chair: Choe, Yoonsuck

**Special Session ss23-1: Complex and memristive networks 1**
Monday, July 13, 10:50AM-12:10PM, Room: Ballroom, Chair: Corinto, Fernando; Lanza, Valentina

10:50AM Class of memristors from cascade of static nonlinear two ports with dynamic one-ports [#15021]

Alon Ascoli, Fernando Corinto and Ronald Tetzlaff

11:10AM Behavioral Model for Simplified Identification of Memristor Parameters [#15336]

Zdenek Kolka, Viera Biolkova, Dalibor Biolek and Jiri Vavra

11:30AM Quantum-Inspired Complex-Valued Multidirectional Associative Memory [#15195]

Naoki Masuyama and ChuKiong Loo

11:50AM Complex-Valued Multilayer Perceptron Learning Using Singular Regions and Search Pruning [#15266]

Seiya Satoh and Ryohei Nakano

**Session Manif1: Manifold learning and dimensionality reduction 1**
Monday, July 13, 10:50AM-12:10PM, Room: Brehon, Chair: Andras, Peter

10:50AM High-Dimensional Function Approximation Using Local Linear Embedding [#15134]

Peter Andras

11:10AM Learning Orthogonal Sparse Representations by using Geodesic Flow Optimization [#15540]

Henry Schuetze, Erhardt Barth and Thomas Martinetz

11:30AM Stochastic Computation of Dominant Eigenvalue and the Law of Total Variance [#15661]

George Georgiou, Kerstin Voigt and Haiyan Qiao

11:50AM Sparsity Analysis of Learned Factors in Multilayer NMF [#15396]

Ievgen Redko and Younes Bennani

**Special Session ss11-2: SS: Biomedical applications**
Monday, July 13, 10:50AM-12:10PM, Room: Park Suite, Chair: Ko, Li-Wei

10:50AM Using Regional Homogeneity from Functional MRI for Diagnosis of ASD among Males [#15413]

Vigneshwaran Subbaraju, Suresh Sundaram, Mahanand Belathur Suresh and Sundararajan Narasimhan

11:10AM An Empirical Mode Decomposition Based Filtering Method for Classification of Motor-Imagery EEG Signals for Enhancing Brain-Computer Interface [#15679]

Pramod Gaur, Ram Bilas Pachori, Hui Wang and Girijesh Prasad

11:30AM Learning with Covariate Shift-Detection and Adaptation in Non-Stationary Environments: Application to Brain-Computer Interface [#15664]

Haider Raza, Hubert Cecotti, Yuhua Li and Girijesh Prasad

11:50AM Single Channel Wireless EEG Device for Real-Time Fatigue Level Detection [#15767]

Li-Wei Ko, Wei-Kai Lai, Wei-Gang Liang, Chun-Hsiang Chuang, Shao-Wei Lu, Yi-Chen Lu, Tien-Yang Hsiung, Hsu-Hsuan Wu and Chin-Teng Lin

**Session Dyn1: Neurodynamics 1**
Monday, July 13, 10:50AM-12:10PM, Room: Mangerton, Chair: Guo, Wentao

10:50AM Parallel Algorithms for a Neurodynamic Optimization System Realized on GPU and Applied to Recovering Compressively Sensed Signals [#15462]

Xiaodan Zhu and Chengan Guo

11:10AM Effect of Associative Rules on the Dynamics of Conceptual Combination in a Neurodynamical Model [#15832]

Sarjoun Doumit and Ali Minai

11:30AM Error Bound Analysis of Policy Iteration Based Approximate Dynamic Programming for Deterministic Discrete-time Nonlinear Systems [#15717]

Wentao Guo, Feng Liu, Jennie Si, Shengwei Mei and Rui Li

11:50AM Predictive Event-Triggered Control based on Heuristic Dynamic Programming for Nonlinear Continuous-Time Systems [#15812]

Lu Dong, Xiangnan Zhong, Changyin Sun and Haibo He

**Session Emotion: Emotion and motivation**
Monday, July 13, 11:30AM-12:10PM, Room: Auditorium, Chair: Perlovsky, Leonid

11:30AM Artificial Motivations based on Drive-Reduction Theory in Self-Referential Model-Building Control Systems [#15499]

Moritz Schneider and Juergen Adamy

11:50AM Towards a Neural Model of Bonding in Self-Attachment [#15393]

David Cittern and Abbas Edalat

**Session Finance1: Financial and commercial applications**
Monday, July 13, 1:30PM-2:30PM, Room: Auditorium, Chair: Gashler, Michael

1:30PM Investing in Emerging Markets Using Neural Networks and Particle Swarm Optimisation [#15710]

Pascal Khoury and Denise Gorse

1:50PM Meta-Learning to Choose the Level of Analysis in Nested Data: A Case Study on Error Detection in Foreign Trade Statistics [#15546]

Mohammad Nozari Zarmehri and Carlos Soares

2:10PM A Hybrid Latent Variable Neural Network Model for Item Recommendation [#15058]

Michael Smith, Michael Gashler and Tony Martinez

**Session EEGMEG1: EEG/MEG analysis and applications 1**
Monday, July 13, 1:30PM-2:30PM, Room: Ballroom, Chair: Diwakar, Shyam

1:30PM Adaptive Parameterized AdaBoost Algorithm with Application in EEG Motor Imagery Classification [#15801]

Pratyusha Das, Arup Kumar Sadhu, Amit Konar, Basabdatta Sen Bhattacharya and Atulya K. Nagar

1:50PM Exploiting Point Source Approximation on Detailed Neuronal Models to Reconstruct Single Neuron Electric Field and Population LFP [#15474]

Harilal Parasuram, Bipin Nair, Giovanni Naldi, Egidio D'Angelo and Shyam Diwakar

2:10PM EEG Signal Analysis for BCI Application using Fuzzy System [#15455]

Thanh Nguyen, Saeid Nahavandi, Abbas Khosravi, Douglas Creighton and Imali Hettiarachchi

**Session Manif2: Manifold learning and dimensionality reduction 2**
Monday, July 13, 1:30PM-2:30PM, Room: Brehon, Chair: Wang, Xiaoping

1:30PM Improved Manifold Learning with Competitive Hebbian Rule [#15798]

Qiang Gan, Furao Shen and Jinxi Zhao

1:50PM Robust Semi-supervised Nonnegative Matrix Factorization [#15222]

Jing Wang, Feng Tian, Chang Hong Liu and Xiao Wang

2:10PM Correntropy Supervised Non-negative Matrix Factorization [#15508]

Zhang Wenju, Guan Naiyang, Tao Dacheng, Mao Bin, Huang Xuhui and Luo Zhigang

**Session Sampling: Sampling**
Monday, July 13, 1:30PM-2:30PM, Room: Park Suite, Chair: Carvalho, Andre

1:30PM From Cutting Planes Algorithms to Compression Schemes and Active Learning [#15601]

Ugo Louche and Liva Ralaivola

1:50PM An Evolutionary Sampling Approach for Classification with Imbalanced Data [#15687]

Everlandio Fernandes, Andre Carvalho and Andre Coelho

2:10PM Lattice point sets for efficient kernel smoothing models [#15285]

Cristiano Cervellera, Mauro Gaggero, Danilo Maccio' and Roberto Marcialis

**Session Dyn2: Neurodynamics 2**
Monday, July 13, 1:30PM-2:30PM, Room: Mangerton, Chair: Guo, Chengan; Guo; Wentao

1:30PM Neural network observer-based optimal control for unknown nonlinear systems with control constraints [#15459]

Yuzhu Huang and Hongde Jiang

1:50PM A Neurodynamic Optimization Approach to Synthesis of Linear Systems with Fault Detection via Robust Pole Assignment [#15159]

Xinyi Le and Jun Wang

2:10PM Spatio-temporal Map Formation Based on a Potential Function [#15187]

Prayag Gowgi and Shayan Garani

**Session Finance2: Financial and commercial applications**
Monday, July 13, 2:40PM-4:20PM, Room: Auditorium, Chair: Quek, Chai

2:40PM Financial Data Analysis Using The Informational Energy Unilateral Dependency Measure [#15653]

Angel Cataron, Razvan Andonie and Yvonne Chueh

3:00PM Three-MLP Ensemble Re-RX Algorithm and Recent Classifiers for Credit-Risk Evaluation [#15166]

Yoichi Hayashi, Yuki Tanaka, Shonosuke Yukita, Satoshi Nakano and Guido Bologna

3:20PM Case-based Reasoning Combined with Neural Networks for Credit Risk Analysis [#15660]

Cesar Silva, Germano Vasconcelos, Gabriel Silva and Hadautho Barros

3:40PM A PAA-PSO technique for investment strategies in the financial market [#15580]

Victor Souza, Rodrigo Brasileiro and Adriano Oliveira

4:00PM trading model: Self Reorganizing Fuzzy Associative Machine - forecasted MACD-Historgram (SeroFAM-fMACDH) [#15426]

Javan Tan, WeiGui jair Zhou and Chai Quek

**Special Session ss23-2: Complex and memristive networks 2**
Monday, July 13, 2:40PM-4:20PM, Room: Ballroom, Chair: Hardy, Leon

2:40PM Controllability of Multi-level States in Memristive Device Models using a Transistor as Current Compliance during SET Operation [#15667]

Anne Siemon, Stephan Menzel, Rainer Waser and Eike Linn

3:00PM Architecture and Simulation of a Hybrid Memristive Multiplier Network using Redundant Number Representation [#15725]

Dietmar Fey and Jonathan Martschinke

3:20PM An application of neurohydrodynamics to a Hopfield neural network [#15121]

Leon Hardy

3:40PM Gradient-descent-based learning in memristive crossbar arrays [#15548]

Manu V Nair and Piotr Dudek

4:00PM Mode-Locking in a Network of Kuramoto-like Oscillators [#15645]

Eugene Koskin, Dimitri Galayko, Orla Feely and Elena Blokhina

**Session DeepNN1: Deep neural networks 1**
Monday, July 13, 2:40PM-4:20PM, Room: Brehon, Chair: Louis, Sushil

2:40PM Deep Convolutional Neural Networks as Generic Feature Extractors [#15583]

Lars Hertel, Erhardt Barth, Thomas Kaester and Thomas Martinetz

3:00PM DeepSign: Deep Learning for Automatic Malware Signature Generation and Classification [#15763]

Omid E. David and Nathan S. Netanyahu

3:20PM Deep Learning Using Partitioned Data Vectors [#15301]

Ben Mitchell, Hasari Tosun and John Sheppard

3:40PM Face Recognition in Unconstrained Environments A Deep Architecture on A Small Training Set [#15744]

Mohammad Taghi Saffar, Banafsheh Rekabdar, Sushil Louis and Mircea Nicolescu

4:00PM Exploring Autoencoders for Unsupervised Feature Selection [#15172]

Chandra B. and Rajesh K. Sharma

**Special Session ss13: Emerging Methodologies for Big Data Integration**
Monday, July 13, 2:40PM-4:20PM, Room: Park Suite, Chair: Tagliaferri, Roberto

2:40PM An Automated String-Based Approach to White Matter Fiber-Bundles Clustering [#15386]

Francesco Cauteruccio, Claudio Stamile, Giorgio Terracina, Domenico Ursino and Dominique Sappey-Marinier

3:00PM Modelling Absence Epilepsy Seizure Data in the NeuCube Evolving Spiking Neural Network Architecture [#15691]

Elisa Capecci, Josafath I. Espinosa Ramos, Nadia Mammone, Nikola Kasabov, Jonas Duun-Henriksen, Troels Wesenberg Kjaer, Maurizio Campolo, Fabio La Foresta and Francesco C. Morabito

3:20PM Learning Vector Quantization and Permutation Entropy to Analyse Epileptic Electroencephalography [#15483]

Nadia Mammone, Jonas Duun-Henriksen, Troels W. Kjaer, Maurizio Campolo, Fabio La Foresta and Francesco C. Morabito

3:40PM Biomedical data integration and ontology-driven multi-facets visualization [#15182]

Carmen De Maio, Giuseppe Fenza, Vincenzo Loia and Mimmo Parente

4:00PM Multi omic oscillations in bacterial pathways [#15831]

Francesco Bardozzo, Pietro Lio and Roberto Tagliaferri

**Session Neurosc2: Neuroscience 2: neurons, synapses, and circuits**
Monday, July 13, 2:40PM-4:20PM, Room: Mangerton, Chair: Salles Chevitarese, Daniel

2:40PM Influence of the refractory period on neural networks based on the recognition of neural signatures [#15067]

Jose Luis Carrillo_Medina and Roberto Latorre

3:00PM Synaptic amplification by axo-shaft synapses in a pyramidal neuron model [#15114]

Youwei Zheng and Lars Schwabe

3:20PM Connectivity estimation of neural networks using a spike train kernel [#15245]

Taro Tezuka and Christophe Claramunt

3:40PM Introduction to CircuitML: Modeling Local Processing Units in the Drosophila Brain [#15153]

Daniel Salles Chevitarese, Dilza Szwarcman and Marley Vellasco

4:00PM Modulation Ratio of Layer 2/3 Cells in Primary Visual Cortex: A Model Based Study [#15161]

Dhanaraj Kakkanattu Jagalchandran and Basabi Bhaumik

**Session Temporal: Time and temporal processes**
Monday, July 13, 4:40PM-6:20PM, Room: Auditorium, Chair: Maniadakis, Michail

4:40PM Distal Dendrite Feedback in Hierarchical Temporal Memory [#15102]

Adam Kneller and John Thornton

5:00PM Artificial Agents Perceiving and Processing Time [#15644]

Michail Maniadakis and Panos Trahanias

5:20PM Learning Joint Representations for Order and Timing of Perceptual-Motor Sequences: a Dynamic Neural Field Approach [#15631]

Weronika Wojtak, Flora Ferreira, Wolfram Erlhagen and Estela Bicho

5:40PM Duration and Interval Hidden Markov Model for Sequential Data Analysis [#15752]

Hiromi Narimatsu and Hiroyuki Kasai

6:00PM Product Reservoir Computing: Time-Series Computation with Multiplicative Neurons [#15261]

Alireza Goudarzi, Shabani Alireza and Stefanovic Darko

**Session EEGMEG2: EEG/MEG analysis and applications 2**
Monday, July 13, 4:40PM-6:20PM, Room: Ballroom, Chair: Prasad, Girijesh

4:40PM Single-trial detection of realistic images with magnetoencephalography [#15711]

Hubert Cecotti and Girijesh Prasad

5:00PM A Hands Free Browser Using EEG and Voice Inputs [#15148]

Joy Bose, Singhai Amit, Trisal Ankur, Keshav Vinod and Dubey Utkarsh

5:20PM Reliable Seizure Prediction from EEG Data [#15062]

Vladimir Cherkassky, Brandon Veber, Jieun Lee, Gregory Worrell, Benjamin Brinkmann, Ned Patterson and Shiao Han-Tai

5:40PM Reward-based online learning in non-stationary environments: adapting a P300-speller with a “Backspace” key [#15588]

Emmanuel Dauce, Timothee Proix and Liva Ralaivola

6:00PM EEG-based Biometric Identification Using Local Probability Centers [#15093]

Chengsheng Mao, Bin Hu, Manman Wang and Moore Philip

**Session DeepNN2: Deep neural networks 2**
Monday, July 13, 4:40PM-6:20PM, Room: Brehon, Chair: Lee, Minho

4:40PM Deep Convolutional Network Neocognitron: Improved Interpolating-Vector [#15345]

Kunihiko Fukushima and Hayaru Shouno

5:00PM A novel deep learning by combining discriminative model with generative model [#15449]

Sangwook Kim, Jixiang Shen and Minho Lee

5:20PM Direct Conversion from Facial Myoelectric Signals to Speech using Deep Neural Networks [#15196]

Lorenz Diener, Matthias Janke and Tanja Schultz

5:40PM Resource-Constrained Classification Using a Cascade of Neural Network Layers [#15467]

Sam Leroux, Steven Bohez, Tim Verbelen, Bert Vankeirsbilck, Pieter Simoens and Bart Dhoedt

6:00PM Multi-Column Deep Neural Networks for Offline Handwritten Chinese Character Classification [#15350]

Dan Ciresan and Ueli Meier

**Special Session ss26: Optimizing Neural Networks Using Evolutionary Computation and Swarm Intelligence**
Monday, July 13, 4:40PM-6:20PM, Room: Park Suite, Chair: Yeh, Wei-Chang

4:40PM Solving reliability redundancy allocation problems with orthogonal simplified swarm optimization [#15220]

Wei-Chang Yeh, Yun-Zhi Jiang*, Vera Yuk Ying Chung and Xiangjian He

5:00PM Analysis and Evaluation of Smartphone-based Human Activity Recognition Using a Neural Network Approach [#15315]

Yongjin Kwon, Kyuchang Kang and Changseok Bae

5:20PM A Probability-Dynamic Particle Swarm Optimization for Object Tracking [#15348]

Feng Sha, Changseok Bae, Guang Liu, XiMeng Zhao, Yuk Ying Chung, WeiChang Yeh and Xiangjian He

5:40PM Competitive Two-Island Cooperative Co-evolution for Training Feedforward Neural Networks for Pattern Classification Problems [#15103]

Rohitash Chandra and Gary Wong

6:00PM Design Static Var Compensator Controller Using Artificial Neural Network Optimized By Modify Grey Wolf Optimization [#15614]

Al-Attar Mohamed, Ahmed El-Gaafary, Yahia Mohamed and Ashraf Hemeida

**Session Neurosc3: Neuroscience 3: behavior, memory, and motivation**
Monday, July 13, 4:40PM-6:20PM, Room: Mangerton, Chair: Mandali, Alekhya

4:40PM Spike-Timing Neuronal Modelling of Forgetting in Immediate Serial Recall [#15441]

Panagiotis Ioannou, Matthew Casey and Andre Gruning

5:00PM A computational basal ganglia model to assess the role of STN-DBS on Impulsivity in Parkinson's disease [#15029]

Alekhya Mandali and Srinivasa Chakravarthy V

5:20PM A Mushroom Bodies inspired spiking network for classification and sequence learning [#15288]

Paolo Arena, Marco Cali', Luca Patane', Agnese Portera and Roland Strauss

5:40PM Modeling pavlovian conditioning with multiple neuronal populations [#15630]

Maxime Carrere and Frederic Alexandre

6:00PM Comparison of meta-analysis approaches for neuroimaging studies of reward processing: A case study [#15237]

Manisha Chawla and Krishna P. Miyapuram

**Plenary Poster Session Poster1: Poster session 1**
Monday, July 13, 7:30PM-9:30PM, Room: * Poster hall, Chair: Yu, Xiao-Hua (Helen)

P101 Random-Forest-Based Automated Cell Detection in Knife-Edge Scanning Microscope Rat Nissl Data [#15830]

Shashwat Lal Das, John Keyser and Yoonsuck Choe

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## Tuesday, July 14, 2015

**Plenary Talk Plenary2: Plenary session: Mario Polycarpou**
Tuesday, July 14, 8:00AM-9:00AM, Room: Auditorium, Chair: Wunsch, Donald C.

**Session Social: Social media analysis**
Tuesday, July 14, 9:10AM-10:30AM, Room: Auditorium, Chair: King, Irwin

9:10AM Discovery of Localized Spatio-Temporal Patterns from Location-based SNS by Clustering Users [#15460]

Ken-ichiro Nishioka, Yoshitatsu Matsuda and Kazunori Yamaguchi

9:30AM Online Diffusion Source Detection in Social Networks [#15263]

Haishuai Wang, Peng Zhang, Ling Chen, Huan Liu and Chengqi Zhang

9:50AM Collaborative Filtering via Co-Factorization of Individuals and Groups [#15435]

Yihai Huang and James Kwok

10:10AM Group Buying in Social Coupon: Myths or Facts [#15827]

Yuanyuan Man, Mantian Hu and Irwin King

**Special Session ss32-1: Autonomous Machine Learning for Cyber-Physical Systems 1**
Tuesday, July 14, 9:10AM-10:30AM, Room: Ballroom, Chair: Ozawa, Seiichi

9:10AM A Federated Network Online Network Traffics Analysis Engine for Cybersecurity [#15415]

Shaoning Pang, Yiming Peng, Tao Ban, Daisuke Inoue and Abdolhossein Sarrafzadeh

9:30AM Interactive Online Learning for Obstacle Classification on a Mobile Robot [#15478]

Viktor Losing, Barbara Hammer and Heiko Wersing

9:50AM A Study on Association Rule Mining of Darknet Big Data [#15768]

Tao Ban, Masashi Eto, Shanqing Guo, Daisuke Inoue, Koji Nakao and Runhe Huang

10:10AM Human Intention Understanding Based On Object Affordance and Action Classification [#15445]

Zhibin Yu, Sangwook Kim, Rammohan Mallipeddi and Minho Lee

**Special Session ss17: Autonomous Learning from Big Data**
Tuesday, July 14, 9:10AM-10:30AM, Room: Brehon, Chair: Angelov, Plamen

9:10AM Typicality Distribution Function - A New Densitybased Data Analytics Tool [#15244]

Plamen Angelov

9:30AM Evolving Clustering, Classification and Regression with TEDA [#15366]

Dmitry Kangin and Plamen Angelov

9:50AM Data Mapping by Restricted Boltzmann Machines for Social Circles Detection [#15542]

Jesus Alonso, Roberto Paredes and Paolo Rosso

10:10AM Automated online feature selection and learning from high-dimensional streaming data using an ensemble of Kohonen neurons [#15001]

Asim Roy

**Session Theory1: Neural networks theory 1**
Tuesday, July 14, 9:10AM-10:30AM, Room: Park Suite, Chair: Siegelmann, Hava

9:10AM Lie Algebra-Valued Neural Networks [#15723]

Calin-Adrian Popa

9:30AM Implementation of Universal Turing Machines via Fixed-precision Neurons [#15833]

Nicholas Hobbs and Hava Siegelmann

9:50AM An Analysis of Dynamic Cortex Memory Networks [#15678]

Sebastian Otte, Marcus Liwicki and Andreas Zell

10:10AM The MC-ELM: Learning an ELM-like network with Minimum VC dimension [#15555]

Jayadeva Dr, Sumit Soman and Amit Bhaya

**Session Predict: Prediction and forecasting**
Tuesday, July 14, 9:10AM-10:30AM, Room: Mangerton, Chair: Blumenstein, Michael

9:10AM Forecasting Solar Power Generated by Grid Connected PV Systems Using Ensembles of Neural Networks [#15429]

Mashud Rana, Irena Koprinska and Vassilios Agelidis

9:30AM Towards robust flood forecasts using neural networks [#15464]

Seyyed Adel Alavi Fazel, Hamid Mirfenderesk, Tomlinson Rodger and Michael Blumenstein

9:50AM On the Method for Data Streams Aggregation to Predict Shoppers Loyalty [#15311]

Vladimir Nikulin

10:10AM A Configurable Deep Network for High-Dimensional Clinical Trial Analysis [#15811]

James O' Donoghue, Mark Roantree and Martin Van Boxtel

**Panel Session Invited2: Invited Speaker: Vladimir Cherkassky**
Tuesday, July 14, 10:50AM-11:30AM, Room: Auditorium, Chair: Morabito, Francesco Carlo

**Special Session ss32-2: Autonomous Machine Learning for Cyber-Physical Systems 2**
Tuesday, July 14, 10:50AM-12:10PM, Room: Ballroom, Chair: Liu, Derong

10:50AM Incremental learning on a budget and a quick calculation method using a tree-search algorithm [#15746]

Akihisa Kato, Hirohito Kawahara and Yamauchi Koichiro

11:10AM An Autonomous Online Malicious Spam Mail Detection System Using Extended RBF Network [#15785]

Aminah Ali Siti Hajar, Seiichi Ozawa, Junji Nakazato, Tao Ban and Jumpei Shimamura

11:30AM A Neural Path Integration Mechanism for Adaptive Vector Navigation in Autonomous Agents [#15190]

Dennis Goldschmidt, Sakyasingha Dasgupta, Florentin Woergoetter and Poramate Manoonpong

11:50AM Data-driven Virtual Reference Controller Design for High-order Nonlinear Systems via Neural Network [#15111]

Pengfei Yan, Derong Liu and Ding Wang

**Special Session ss25: Computational Intelligence Applications to Environmental Sustainability and Sustainable Development: Theory and Applications**
Tuesday, July 14, 10:50AM-12:10PM, Room: Brehon, Chair: Tambouratzis, Tatiana; Grozavu, Nistor

10:50AM Adaptive Memetic Algorithm for the Job Shop Scheduling Problem [#15205]

Jakub Nalepa, Marcin Cwiek and Michal Kawulok

11:10AM How reliable is the Environmental Sustainability Index 2005? [#15742]

Tatiana Tambouratzis

11:30AM Approximate Dynamic Programming for Control of a Residential Water Heater [#15654]

Matthew Motoki, Monica Umeda, Matthias Fripp and Tony Kuh

11:50AM A RBF Neural Network applied to predict soil Field Capacity and Permanent Wilting Point at Brazilian Coast [#15507]

Giorgia Carvalho, Diego Brandao, Diego Haddad, Vinicius Forte and Marcos Ceddia

**Session Theory2: Neural networks theory 2**
Tuesday, July 14, 10:50AM-12:10PM, Room: Park Suite, Chair: Farkas, Igor

10:50AM A Method for Finding Similarity between Multi-Layer Perceptrons by Forward Bipartite Alignment [#15698]

Stephen Ashmore and Michael Gashler

11:10AM Regularity and Randomness in Modular Network Structures for Neural Associative Memories [#15792]

Gouhei Tanaka, Toshiyuki Yamane, Daiju Nakano, Ryosho Nakane and Yasunao Katayama

11:30AM On the Performance of Quaternionic Bidirectional Auto-Associative Memory [#15594]

Toshifumi Minemoto, Teijiro Isokawa, Masaki Kobayashi, Haruhiko Nishimura and Nobuyuki Matsui

11:50AM Computational Analysis of the Bidirectional Activation-based Learning in Autoencoder Task [#15322]

Peter Csiba and Igor Farkas

**Session Ensemble: Ensemble learning**
Tuesday, July 14, 10:50AM-12:10PM, Room: Mangerton, Chair: Cavalcanti, George

10:50AM Adaptive Skew-Sensitive Fusion of Ensembles and their Application to Face Re-Identification [#15234]

Miguel De-la-Torre, Eric Granger and Robert Sabourin

11:10AM A Bootstrap-Based Iterative Selection for Ensemble Generation [#15602]

Dayvid Oliveira, Thyago Porpino, George Cavalcanti and Tsang Ren

11:30AM An Analysis of Diversity Measures for the dynamic design of ensemble of classifiers [#15823]

Jose Lustosa-Filho, Anne Canuto and Joao Carlo Xavier Junior

11:50AM A Binary Ensemble Classifier for High-Frequency Trading [#15468]

Everton Silva, Humberto Brandao, Douglas Castilho and Adriano Pereira

**Session SVM1: Support Vector Machines 1**
Tuesday, July 14, 11:30AM-12:10PM, Room: Auditorium, Chair: Anguita, Davide

11:30AM Shrinkage Learning to Improve SVM with Hints [#15209]

Luca Oneto, Alessandro Ghio, Sandro Ridella and Davide Anguita

11:50AM A Transductive SVM with Quasi-linear Kernel Based on Cluster Assumption for Semi-Supervised Classification [#15302]

Bo Zhou, Di Fu, Chao Dong and Jinglu Hu

**Plenary Talk Plenary3: Plenary session: Vincenzo Piuri**
Tuesday, July 14, 1:30PM-2:30PM, Room: Auditorium, Chair: Prokhorov, Danil

**Session SVM2 : Support Vector Machines 2**
Tuesday, July 14, 2:40PM-4:20PM, Room: Auditorium, Chair: Carvalho, Andre

2:40PM Optimizing Working Sets for Training Support Vector Regressors by Newton's Method [#15028]

Shigeo Abe

3:00PM Graph-based Semi-supervised Support Vector Data Description for Novelty Detection [#15419]

Phuong Duong, Van Nguyen, Mi Dinh, Trung Le, Dat Tran and Wanli Ma

3:20PM To tune or not to tune: recommending when to adjust SVM hyper-parameters via Meta-learning [#15531]

Rafael G. Mantovani, Andre L. D. Rossi, Bernd Bischl, Joaquin Vanschoren and Andre C. P. L. F. Carvalho

3:40PM Multiple SVM-RFE for Multi-class Gene Selection on DNA Microarray Data [#15216]

Li Zhang and XiaoJuan Huang

4:00PM Improving ESVM with Generalized Cross-Validation [#15053]

Tianshu Feng, Fuzhen Zhuang and Qing He

**Session RNN: Recurrent neural networks**
Tuesday, July 14, 2:40PM-4:20PM, Room: Ballroom, Chair: Hussain, Abir

2:40PM Transient Phenomena Prediction Using Recurrent Neural Networks [#15578]

Jonathan Guerra, Patricia Klotz, Beatrice Laurent and Fabrice Gamboa

3:00PM Computational Capabilities of Recurrent Neural Networks Based on their Attractor Dynamics [#15536]

Jeremie Cabessa and Alessandro Villa

3:20PM Recurrent Convolutional Neural Networks for Object-Class Segmentation of RGB-D Video [#15773]

Mircea Serban Pavel, Hannes Schulz and Sven Behnke

3:40PM Discriminative Learning and Inference in the Recurrent Temporal RBM for Melody Modelling [#15593]

Srikanth Cherla, Son Tran, Artur d'Avila Garcez and Tillman Weyde

4:00PM On the Existence of Hopfield Neural Networks: Synthesis of Hopfield Type Associative Memories [#15010]

Garimella Rama Murthy and Moncef Gabbouj

**Special Session ss03: Cognition and Development**
Tuesday, July 14, 2:40PM-4:20PM, Room: Brehon, Chair: Di Nuovo, Alessandro

2:40PM Deterministic chaos in mobile robots [#15740]

Federico Da Rold

3:00PM Neural-Symbolic Monitoring and Adaptation [#15627]

Alan Perotti, Artur d'Avila Garcez and Guido Boella

3:20PM Symbolic associations in neural network activations: representations in the emergence of communication [#15616]

Emerson Oliveira and Angelo Loula

3:40PM Brains as Naturally Emerging Turing Machines [#15807]

Juyang Weng

4:00PM Data-point and Feature Selection of Motor Imagery EEG Signals for Neural Classification of Cognitive Tasks in Car-Driving [#15797]

Anuradha Saha, Amit Konar, Pratyusha Das, Basabdatta Sen Bhattacharya and Atulya K. Nagar

**Session Theory3: Neural networks theory 3**
Tuesday, July 14, 2:40PM-4:20PM, Room: Park Suite, Chair: Battiti, Roberto

2:40PM Stochastic Local Search for Direct Training of Threshold Networks [#15699]

Mauro Brunato and Roberto Battiti

3:00PM Reliable estimation of a neural network's domain of validity through interval analysis based inversion [#15734]

Stavros Adam, Dimitrios Karras, George Magoulas and Michael Vrahatis

3:20PM A Bounded Neural Network for Open Set Recognition [#15579]

Douglas Cardoso, Felipe Franca and Joao Gama

3:40PM The Generalized Group Lasso [#15480]

Carlos M. Alaiz and Jose R. Dorronsoro

4:00PM Quasi-Newton Learning Methods for Complex-Valued Neural Networks [#15258]

Calin-Adrian Popa

**Special Session ss36: Ensemble Systems and Machine Learning**
Tuesday, July 14, 2:40PM-4:20PM, Room: Mangerton, Chair: Ludermir, Teresa

2:40PM META-DES.H: a dynamic ensemble selection technique using meta-learning and a dynamic weighting approach [#15456]

Rafael M. O. Cruz, Robert Sabourin and George D. C. Cavalcanti

3:00PM Differential Evolution and Meta-Learning for Dynamic Ensemble of Neural Network Classifiers [#15400]

Tiago Lima and Teresa Ludermir

3:20PM Multi-Privacy Biometric Protection Scheme using Ensemble Systems [#15547]

Damasceno Marcelo, Canuto Anne and Poh Norman

3:40PM Growing Hierarchical Trees for Data Stream Clustering and Visualization [#15185]

Nhat-Quang Doan, Mohammed Ghesmoune, Hanane Azzag and Mustapha Lebbah

4:00PM Probabilistic Relational Models with Clustering Uncertainty [#15113]

Anthony Coutant, Philippe Leray and Hoel Le Capitaine

**Session SVM3: Support Vector Machines 3**
Tuesday, July 14, 4:40PM-6:20PM, Room: Auditorium, Chair: Zhang, Li

4:40PM L3-SVM: a LifeLong Learning Method for SVM [#15149]

Youlu Xing, Furao Shen, Chaomin Luo and Jinxi Zhao

5:00PM A fast approximation algorithm for 1-norm SVM with squared loss [#15061]

Li Zhang, Weida Zhou, Zhao Zhang and Jiwen Yang

5:20PM A PARTAN-Accelerated Frank-Wolfe Algorithm for Large-Scale SVM Classification [#15193]

Emanuele Frandi, Ricardo Nanculef and Johan A. K. Suykens

5:40PM Least Square Support Vector Machine for Large-scale Dataset [#15430]

Khanh Nguyen, Trung Le, Vinh Lai, Duy Nguyen, Dat Tran and Wanli Ma

6:00PM Support Vector Machines and Strictly Positive Definite Kernel: The Regularization Hyperparameter is More Important than the Kernel Hyperparameters [#15210]

Luca Oneto, Alessandro Ghio, Sandro Ridella and Davide Anguita

**Special Session ss29: Modeling and Forecasting Financial and Commodity Markets by Neural Networks**
Tuesday, July 14, 4:40PM-6:20PM, Room: Ballroom, Chair: Panella, Massimo; Girijesh, Prasad

4:40PM A Graphical Model Framework for Stock Portfolio Construction with Application to a Neural Network Based Trading Strategy [#15124]

Mininder Sethi and Philip Treleaven

5:00PM Maximum Length Weighted Nearest Neighbor Approach for Electricity Load Forecasting [#15753]

Tommaso Colombo, Irena Koprinska and Massimo Panella

5:20PM A Prediction Model for High-Frequency Financial Time Series [#15304]

Ricardo de A. Araujo, Adriano L. I. Oliveira and Silvio R. de L. Meira

5:40PM Kanban Cell Neuron Network: Stock Trading System (KCNSTS) [#15130]

Colin James III

6:00PM Prediction of Solar Cycle 24 Using a Connectionist Model of the Emotional System [#15808]

Mahboobeh Parsapoor, Mahboobeh Parsapoor, Urban Blistrup and Bertil Svensson

**Special Session ss27: Models of Cognitive-Emotional Interactions**
Tuesday, July 14, 4:40PM-6:20PM, Room: Brehon, Chair: Levine, Daniel

4:40PM Modelling Emotional Attachment: an Integrative Framework for Architectures and Scenarios. [#15232]

Dean Petters and Everett Waters

5:00PM Mystery in Experimental Psychology, Aesthetic Emotions [#15296]

Leonid Perlovsky

5:20PM A Minimal Architecture for General Cognition [#15673]

Michael S. Gashler, Zachariah Kindle and Michael R. Smith

5:40PM Introduction to Self-attachment and its Neural Basis [#15714]

Abbas Edalat

6:00PM A Neural Network Model of Decisions on the Asian Disease Problem [#15292]

Bakur AlQaudi, Daniel Levine and Frank Lewis

**Session Language: Natural language processing**
Tuesday, July 14, 4:40PM-6:20PM, Room: Park Suite, Chair: Hagiwara, Masafumi

4:40PM A Cascade of Linguistic CMAC Neural Networks for Decision Making [#15198]

Hongmei He, Zhenhuan Zhu, Ashutosh Tiwari and Mills Andrew

5:00PM Integrating Word Embeddings and Traditional NLP Features to Measure Textual Entailment and Semantic Relatedness of Sentence Pairs [#15275]

Jiang Zhao, Man Lan, Zheng-Yu Niu and Yue Lu

5:20PM Is Learning by Reading a Book Better Than Watching a Movie? A Computational Analysis of Semantic Concept Network Growth During Text and Multimedia Comprehension [#15688]

Naser Al Madi and Javed Khan

5:40PM A Natural Language Processing Neural Network Comprehending English [#15309]

Yuanzhi Ke and Masafumi Hagiwara

6:00PM Generating Image Description by Modeling Spatial Context of an Image [#15541]

Kan Li and Lin Bai

**Session BioApp: Biomedical applications**
Tuesday, July 14, 4:40PM-6:20PM, Room: Mangerton, Chair: Li, Kan

4:40PM Hierarchical Classification of Gene Ontology-based Protein Functions with Neural Networks [#15290]

Ricardo Cerri, Rodrigo Barros and Andre Carvalho

5:00PM Hierarchical Dirichlet Process Hidden Markov Model for Unsupervised Bioacoustic Analysis [#15663]

Marius Bartcus, Faicel Chamroukhi and Hervé Glotin

5:20PM New Insights into The Landscape Relationships of Host Response to Bacterial Pathogens [#15206]

Xiaoyao Yin, Lu Han, Naiyang Guan, Yun Bai, Cong Niu, Hui Bai, Xiaochen Bo and Zhigang Luo

5:40PM Diagnosis of Pneumonia From Sounds Collected Using Low Cost Cell Phones [#15044]

Insu Song

6:00PM An insight on complexity measures and classification in microarray data [#15018]

Veronica Bolon-Canedo, Laura Moran-Fernandez and Amparo Alonso-Betanzos

**Plenary Poster Session Poster2: Poster session 2**
Tuesday, July 14, 7:30PM-9:30PM, Room: * Poster hall, Chair: Yu, Xiao-Hua (Helen)

P301 sEMG-Based Torque Estimation for Robot-Assisted Lower Limb Rehabilitation [#15257]

Long Peng, Zeng-Guang Hou, Nikola Kasabov, Jin Hu, Liang Peng and Wei-Qun Wang

P302 Simplified and Gradual Information Control for Improving Generalization Performance of Multi-Layered Neural Networks [#15294]

Ryotaro Kamimura

P303 State Preserving Extreme Learning Machine for Face Recognition [#15724]

Md. Zahangir Alom, Paheding Sidike, Vijayan Asari and Tarek Taha

P304 A Non-Sigmoidal Activation Function for Feedforward Artificial Neural Networks [#15247]

Pravin Chandra, Udayan Ghose and Apoorvi Sood

P305 A Memetic Algorithm Based Extreme Learning Machine for Classification [#15157]

Yongshan Zhang, Zhihua Cai, Jia Wu, Xinxin Wang and Xiaobo Liu

P306 A Pragmatic Approach to Multi-Class Classification [#15697]

Thomas Kopinski, Stephane Magand, Uwe Handmann and Alexander Gepperth

P307 The Neural-SIFT Feature Descriptor for Visual Vocabulary Object Recognition [#15551]

Sybren Jansen, Amirhosein Shantia and Marco Wiering

P308 Morphological Extreme Learning Machines Applied to Detect and Classify Masses in Mammograms [#15705]

Washington Azevedo, Sidney Lima, Isabela Fernandes, Arthur Rocha, Filipe Cordeiro, Abel Silva-Filho and Wellington Santos

P309 Proof of Hidden Node Number in MLP and Experiments on Well Log Data Inversion [#15255]

Kou-Yuan Huang, Liang-Chi Shen and Jiun-Der You

P310 Fast Convergence of Extended Rademacher Complexity Bounds [#15211]

Luca Oneto, Alessandro Ghio, Sandro Ridella and Davide Anguita

P311 Learning Rule for Associative Memory in Recurrent Neural Networks [#15372]

Theju Jacob and Wesley Snyder

P312 Application of Cooperative Neuro-evolution of Elman Recurrent Networks for a Two-Dimensional Cyclone Track Prediction for the South Pacific Region [#15180]

Rohitash Chandra, Kavina Dayal and Nicholas Rollings

P313 Selective Potentiality Maximization for Input Neuron Selection in Self-Organizing Maps [#15382]

Ryotaro Kamimura and Ryozo Kitajima

P314 Sensor Signal Clustering with Self-Organizing Maps [#15431]

Razvan Popovici and Razvan Andonie

P315 EEG Classification to Determine the Degree of Pleasure Levels in Touch-Perception of Human Subjects [#15642]

Anuradha Saha, Amit Konar, Basabdatta Sen Bhattacharya and Atulya K. Nagar

P316 Generalized Constraint Neural Network Regression Model Subject to Equality Function Constraints [#15335]

Linlin Cao and Baogang Hu

P317 An Efficient Learning Method for RBF Neural Networks [#15685]

Maryam Pazouki, Zijun Wu, Zhixing Yang and Dietmar Moeller

P318 Local Radial Basis Function Network Regressor with Feature Importance Optimization [#15094]

Yu-Ann Chen and Pau-Choo Chung

P319 FIE-FCMAC: A Fuzzy CMAC using Fuzzy Interpolation and Extrapolation [#15265]

WeiGui jair Zhou, Chai Quek and Douglas Leslie Maskell

P320 Bio-inspired hierarchical framework for multi-view face detection and pose estimation [#15570]

Niall McCarroll, Ammar Belatreche, Jim Harkin and Yuhua Li

P321 Arithmetic Computing via Rate Coding in Neural Circuits with Spike-triggered Adaptive Synapses [#15776]

Sushrut Thorat and Bipin Rajendran

P322 Knowledge Extraction Using Probabilistic Reasoning: An Artificial Neural Network Approach [#15364]

Chelsea Dobbins and Paul Fergus

P323 Face Recognition Using Special Neural Networks [#15191]

Ernst Kussul and Tetyana Baydyk

P324 A Label Compression Coding Approach through Maximizing Dependence between Features and Labels for Multi-label Classification [#15226]

Lei Cao and Jianhua Xu

P325 Using Classifier Diversity to Handle Label Noise [#15043]

Michael Smith and Tony Martinez

P326 In-Training and Post-Training Generalization Methods: the case of ppar alpha and ppar gamma agonists [#15410]

Babak Keshavarz-Hedayati, Pan Guangyuan, Ali Jooya and Nikitas Dimopoulos

P327 A Constrained Recursive Least Squares Algorithm For Adaptive Combination of Multiple Models [#15008]

Xia Hong and Yu Gong

P328 Efficient Conformal Regressors using Bagged Neural Nets [#15690]

Ulf Johansson, Cecilia Sonstrod and Henrik Linusson

P329 Learning from Neighborhood for Classification with Local Distribution Characteristics [#15473]

Chengsheng Mao, Bin Hu, Manman Wang and Moore Philip

P330 Real-time occupancy estimation using environmental parameters [#15715]

Mustafa Khalid Masood, Soh Yeng Chai and Victor Chang

P331 Class-Preserving Manifold Learning for Detection and Classification [#15775]

Puoya Tabaghi and Mahmood R. Azimi-Sadjadi

P332 Incremental Probabilistic Classification Vector Machine with linear costs [#15145]

Frank-Michael Schleif, Huanhuan Chen and Peter Tino

P333 Epidemiological Time Series Forecasted with a Time Varying Auto-Adaptive Neural Network [#15387]

Antonio Ballarin, Simona Gervasi, Kristian A. Gervasi Vidal, Marco M Ballarin, Marco Di Francesco, Roberto Tonelli and Giovanni Ballarin

P334 A Learning Scheme Based on Similarity Functions for Affective Common-Sense Reasoning. [#15517]

Federica Bisio, Paolo Gastaldo, Rodolfo Zunino and Erik Cambria

P335 Some Further Evidence about Magnification and Shape in Neural Gas [#15395]

Giacomo Parigi, Andrea Pedrini and Marco Piastra

P336 Stock Price Prediction based on Stock-Specific and Sub-Industry-Specific News Articles [#15353]

Yauheniya Shynkevich, T. M. McGinnity, Sonya Coleman and Ammar Belatreche

P337 Low Rank Sequential Subspace Clustering [#15063]

Yi Guo, Junbin Gao, Feng Li, Stephen Tierney and Ming Yin

P338 Meta-Path based Nonnegative Matrix Factorization for Clustering on Multi-type Relational Data [#15371]

Yangyang Zhao, Zhengya Sun, Changsheng Xu and Hongwei Hao

P339 Applying the Coral Reefs Optimization Algorithm to Clustering Problems [#15815]

Inacio Medeiros, Joao-Carlos Xavier-Junior and Canuto Anne

P340 Online Detection and Modeling of Safety Boundaries for Aerospace Applications using Active Learning and Bayesian Statistics [#15458]

Yuning He

P341 Geometric Approach of Quasi-Linear Kernel Composition for Support Vector Machine [#15158]

Weite Li and Jinglu Hu

P342 Filtering SAR imagery for edge detection using support value transform [#15298]

Li Zhang, Weida Zhou and Bangjun Wang

P343 Using Support Vector Machines and Two Dimensional Discrete Cosine Transform in Speech Automatic Recognition [#15200]

Gracieth Gracieth Cavalcanti Batista and Washington Silva

P344 Multi-Kernel Probability Distribution Regressions [#15432]

Pingping Zhu, Hongchuan Wei, Wenjie Lu and Silvia Ferrari

P345 PAC-Bayes Analysis for Twin Support Vector Machines [#15117]

Xijiong Xie and Shiliang Sun

P346 Face Recognition using Support Vector Machine and Multiscale Directional Image Representation Methods: A comparative study [#15608]

Daniel Costa, Sarajane Peres, Pollyana Mustaro and Clodoaldo Lima

P347 Improved Multi-kernel SVM for Multi-modal and Imbalanced Dialogue Act Classification [#15249]

Yucan Zhou, Xiaowei Cui, Qinghua Hu and Yuan Jia

P348 Solving the Data Imbalance Problem of P300 Detection via Random Under-Sampling Bagging SVMs [#15800]

Xiaofeng Shi, Guoqiang Xu, Furao Shen and Jinxi Zhao

P349 Benchmarking the Semi-Supervised Naive Bayes Classifier [#15558]

Awat Saeed, Gavin Cawley and Anthony Bagnall

P350 Semi-Supervised Min-Max Modular SVM [#15333]

Yan-Ping Wu and Yun Li

P351 Sparse Density Estimation On Multinomial Manifold Combining Local Component Analysis [#15015]

Xia Hong and Junbin Gao

P352 Collaborative Clustering with Heterogeneous Algorithms [#15107]

Jeremie Sublime, Nistor Grozavu, Younes Bennani and Antoine Cornuejols

P353 An Efficient Recognition Algorithm for Restricted Bayesian Networks [#15065]

Yuuji Ichisugi and Naoto Takahashi

P354 Threshold Optimization of Pseudo-Inverse Linear Discriminants Based on Overall Accuracies [#15728]

Tian Tian, Ji Wang and Daqi Gao

P355 An Extended Fuzzy Local Information C-Means Clustering Algorithm [#15020]

Lili Hou, Le Zhang, Qiuying Yang and Ying Wen

P356 Two-layer Mixture of Factor Analyzers with Joint Factor Loading [#15104]

Xi Yang, Huang Kaizhu, Zhang Rui and John Yannis Goulermas

P357 An investigation into the use of subspace methods for face detection [#15584]

Salaheddin Alakkari, Eugene Gath and John James Collins

P358 A Multi-label Feature Selection Algorithm Based on Multi-objective Optimization [#15139]

Jing Yin, Tengfei Tao and Jianhua Xu

P359 Feature Selection using Deep Neural Networks [#15503]

Debaditya Roy, Sri Rama Murty K. and Krishna Mohan C.

P360 Improving Load Forecast Accuracy by Clustering Consumers using Smart Meter Data [#15176]

Abbas Shahzadeh, Abbas Khosravi and Saeid Nahavandi

P361 A Differential Evolution-Based Method for Class-Imbalanced Cost-Sensitive Learning [#15219]

Chen Qiu, Liangxiao Jiang and Ganggang Kong

P362 Effectiveness of Random Search in SVM hyper-parameter tuning [#15557]

Rafael G. Mantovani, Andre L. D. Rossi, Bernd Bischl, Joaquin Vanschoren and Andre C. P. L. F. Carvalho

P363 Face Expression Recognition with a 2-Channel Convolutional Neural Network [#15380]

Dennis Hamester, Pablo Barros and Stefan Wermter

P364 A Combination of Multi-state Activation Functions, Mean-normalisation and Singular Value Decomposition for Learning Deep Neural Networks [#15052]

Chenghao Cai, Dengfeng Ke, Yanyan Xu and Kaile Su

P365 Regularizing Neural Networks with Adaptive Local Drop [#15422]

Binbin Cao, Jianmin Li and Bo Zhang

P366 Forecasting the Weather of Nevada: A Deep Learning Approach [#15760]

Moinul Hossain, Banafsheh Rekabdar, Sushil Louis and Sergiu Dascalu

P367 Analysis of Function of Rectified Linear Unit Used in Deep learning [#15433]

Kazuyuki Hara, Daisuke Saito and Hayaru Shouno

P368 Stochastic Least Squares Learning for Deep Architectures [#15330]

Girish Kumar, Jian Min Sim, Eng Yeow Cheu and Xiaoli Li

P369 Optimized Deep Belief Networks on CUDA GPUs [#15341]

Teng Li, Yong Dou, Jingfei Jiang, Yueqing Wang and Qi Lv

P370 An Empirical Analysis of Different Sparse Penalties for Autoencoder in Unsupervised Feature Learning [#15423]

Nan Jiang, Wenge Rong, Baolin Peng, Yifan Nie and Zhang Xiong

P371 An overview on the Gaussian Fields and Harmonic Functions Method for Semi-supervised Learning [#15308]

Celso Sousa

P372 Evolutionary Adaptive Self-Generating Prototypes for Imbalanced Datasets [#15611]

Dayvid Oliveira, George Cavalcanti, Tsang Ren and Ricardo Silva

P373 A simulator for Freeman K-sets in Java [#15281]

Denis Piazentin and Joao Luis Rosa

P374 Reconstructing fMRI BOLD signals arising from Cerebellar Granule Neurons - Comparing GLM and Balloon Models [#15524]

Chaitanya Medini, Bipin Nair, Giovanni Naldi, Egidio D'Angelo and Shyam Diwakar

P375 EEG Source Localization by Memory Network Analysis of Subjects Engaged in Perceiving Emotions from Facial Expressions [#15615]

Reshma Kar, Amit Konar, Aruna Chakraborty, Basabdatta Sen Bhattacharya and Atulya Nagar

P376 Forecasting Model for Bidding Behavior of Advertisers Based on HMM [#15140]

Long Lili, Dong Hongbin, Pan Yue, Huangfu Li, Gou Naikang and Wang Xingmei

P377 Self-organization of Hippocampal Representations in Large Environments [#15821]

Shuang Liu, Bailu Si and Yang Lin

P378 ANSWER: An Unsupervised Attractor Network Method for Detecting Salient Words in Text Corpora [#15704]

Madhavun Candadai, Aashay Vanarase, Mei Mei and Ali Minai

P379 ASD Detection in Males Using MRI- An Age-group Based Study [#15377]

Vigneshwaran Subbaraju, Suresh Sundaram, Mahanand Belathur Suresh and Sundararajan Narasimhan

P380 Si elegans: Hardware Architecture and Communications Protocol [#15701]

Pedro Machado, Appiah Kofi, Martin McGinnity, John Wade and Martin McGinnity

P381 A Novel Hardware-Efficient Cochlea Model based on Asynchronous Cellular Automaton [#15745]

Masato Izawa and Hiroyuki Torikai

P382 HMAX Model: A Survey [#15576]

Chang Liu and Fuchun Sun

P383 Composer Classification based on Temporal Coding in Adaptive Spiking Neural Networks [#15765]

Chaitanya Prasad Narisetty, Krishnakant Saboo and Bipin Rajendran

P384 Clustering-based gene-subnetwork biomarker identification using gene expression data [#15721]

Narumol Doungpan, Worrawat Engchuan, Asawin Meechai and Jonathan Chan

P385 An HMM-based Gait Comparison on Inertial Sensors: Using Alzheimer's Disease Patients as Examples [#15735]

Wei-Hsin Wang, Hao-Li Wu, Pau-Choo Chung and Ming-Chyi Pai

P386 Short Answer Question Examination using an Automatic Off-line Handwriting Recognition System and a Novel Combined Feature [#15378]

Hemmaphan Suwanwiwat, Umapada Pal and Michael Blumenstein

P387 Recognizing Visual Composite in Real Images [#15361]

Lin Bai, Kan Li and Shuai Jiang

P388 Modelling of a retinal ganglion cell with simple spiking models [#15686]

Philip Vance, Sonya Coleman, Dermot Kerr, Gautham Das and Martin McGinnity

P389 Bag-of-Visual Words for Word-Wise Video Script Identification: A Study [#15511]

Nabin Sharma, Ranju Mandal, Rabi Sharma, Umapada Pal and Michael Blumenstein

P390 An Online Incremental Learning Algorithm For Time Series [#15799]

Haoran Xu, Youlu Xing, Furao Shen and Jinxi Zhao

P391 The Impact of Longstanding Messages in Micro-Blogging Classification [#15650]

Joana Costa, Catarina Silva, Mario Antunes and Bernardete Ribeiro

P392 Interval-valued Symbolic Representation based Method for Off-line Signature Verification [#15354]

Srikanta Pal, Alireza Alaei, Umapada Pal and Michael Blumenstein

P393 Improving Deep Neural Networks Using Softplus Units [#15271]

Hao Zheng, Zhanlei Yang, Jizhong Liang, Yanpeng Li and Wenju Liu

P394 Restoring High Frequency Spectral Envelopes Using Neural Networks for Speech Bandwidth Extension [#15300]

Yu Gu and Zhen-Hua Ling

P395 Indoor Localization by Denoising Autoencoders and Semi-supervised Learning in 3D Simulated Environment [#15629]

Amirhossein Shantia, Rik Timmers, Lambert Schomaker and Marco Wiering

P396 Event-Triggered Adaptive Dynamic Programming for Continuous-time Nonlinear System Using Measured Input-Output Data [#15287]

Xiangnan Zhong, Zhen Ni and Haibo He

P397 Efficient Use of Nadaraya-Watson Models and Low-discrepancy Sequences for Approximate Dynamic Programming [#15399]

Cristiano Cervellera, Mauro Gaggero, Danilo Maccio and Roberto Marcialis

P398 Neurocontrol of Single Shaft Heavy duty Gas Turbine Using Adaptive Dynamic Programming [#15550]

Yuzhu Huang and Hongde Jiang

P399 Music Genre Classification with Self-Organizing Maps and Edit Distance [#15409]

Razvan Popovici and Razvan Andonie

P400 Combining PCA and Multiset CCA for Dimension Reduction when Group ICA is Applied to Decompose Naturalistic fMRI Data [#15636]

Valeri Tsatsishvili, Fengyu Cong, Petri Toiviainen and Tapani Ristaniemi

P401 Scalable Real-time Parking Lot Classification: An Evaluation of Image Features and Supervised Learning Algorithms [#15050]

Marc Tschentscher, Christian Koch, Markus Koenig, Jan Salmen and Marc Schlipsing

P402 Automatic Classification of Drum Sounds with Indefinite Pitch [#15089]

Vinicius Souza, Gustavo Batista and Nilson Souza-Filho

P403 Compressed representation learning for fluid field reconstruction from sparse sensor observations [#15355]

Hongming Zhou, Yeng Chai Soh, Chaoyang Jiang and Xiaoying Wu

P404 An Echo State Network Approach to Structural Health Monitoring [#15505]

Adam Wootton, Charles Day and Peter Haycock

P405 Realistic and Very Fast Simulation of Individual Electricity Consumptions [#15082]

Alexis Bondu and Asma Dachraoui

P406 Echo state networks, artificial neural networks and fuzzy systems models for improve short-term wind speed forecasting [#15589]

Ronaldo Aquino, Ramon Souza, Otoni Nobrega Neto, Milde Lira, Manoel Carvalho and Aida Ferreira

P407 Prediction Interval Estimation for Wind Farm Power Generation Forecasts Using Support Vector Machines [#15563]

Nitin Anand Shrivastava, Abbas Khosravi and Bijaya Ketan Panigrahi

P408 Network-Traffic Anomaly Detection with Incremental Majority Learning [#15428]

Huang Shin-Ying, Yu Fang, Tsaih Rua-Huan and Huang Yennun

P409 Probabilistic Temporal Bilinear Model for Temporal Dynamic Recommender Systems [#15487]

Cheng Luo, Xiongcai Cai and Nipa Chowdhury

P410 A Word Distributed Representation Based Framework for Large-scale Short Text Classification [#15343]

Di Yao, Jingping Bi, Jianhui Huang and Zhu Jin

## Wednesday, July 15, 2015

**Plenary Talk Plenary4: Plenary session: Barak Pearlmutter**
Wednesday, July 15, 8:00AM-9:00AM, Room: Auditorium, Chair: Huang, De-Shuang

**Session Struct: Structures and hierarchies**
Wednesday, July 15, 9:10AM-10:30AM, Room: Auditorium, Chair: Koprinkova-Hristova, Petia

9:10AM Evolving Artificial Neural Networks through L-System and Evolutionary Computation [#15375]

Lidio Mauro Lima De Campos, Roberto Celio Limao De Oliveira and Mauro Roisenberg

9:30AM On effects of IP improvement of ESN reservoirs for reflecting of data structure [#15613]

Petia Koprinkova-Hristova

9:50AM Order-aware exemplars for structuring video sets: Clustering, aligned matching and retrieval by similarity [#15223]

Yasuo Matsuyama, Akihiro Shikano, Hiromichi Iwase and Teruki Horie

10:10AM Improvement of Reliabilities of Regulations using a Hierarchical Structure in a Genetic Network [#15125]

Shuhei Kimura and Mariko Okada-Hatakeyama

**Special Session ss33: Intelligent Vehicle Systems**
Wednesday, July 15, 9:10AM-10:30AM, Room: Ballroom, Chair: Murphy, Yi

9:10AM Approaching Real-World Navigation Using Object Recognition Network [#15643]

Zejia Zheng and Weng Juyang

9:30AM Driver Yawning Detection based on Deep Convolutional Neural Learning and Robust Nose Tracking [#15420]

Weiwei Zhang, Yi Murphey, Tianyu Wang and Qijie Xu

9:50AM Extending Traffic Light Recognition: Efficient Classification of Phase and Pictogram [#15326]

Matthias Michael and Marc Schlipsing

10:10AM Maneuver Segmentation for Autonomous Parking Based on Ensemble Learning [#15389]

Gennaro Notomista and Michael Botsch

**Session Spiking1: Spiking neural networks 1**
Wednesday, July 15, 9:10AM-10:30AM, Room: Brehon, Chair: Dora, Shirin

9:10AM Limits of Coincidence Detector Neurons as Decoders of Polychronous Neuronal Groups Firing Completely [#15427]

Joao Paulo Cerquinho Cajueiro and Joao Ranhel

9:30AM NormAD - Normalized Approximate Descent based Supervised Learning Rule for Spiking Neurons [#15488]

Navin Anwani and Bipin Rajendran

9:50AM CARLsim 3: A User-Friendly and Highly Optimized Library for the Creation of Neurobiologically Detailed Spiking Neural Networks [#15225]

Michael Beyeler, Kristofor Carlson, Ting-Shuo Chou, Nikil Dutt and Jeffrey Krichmar

10:10AM A Two Stage Learning Algorithm for a Growing-Pruning Spiking Neural Network for Pattern Classification Problems [#15454]

Shirin Dora, Sundaram Suresh and Narasimhan Sundararajan

**Session Robot1: Robotics 1: Spatial Cognition and Navigation**
Wednesday, July 15, 9:10AM-10:30AM, Room: Park Suite, Chair: Jung, Seul

9:10AM A Spatial Cognition Model Integrating Grid Cells and Place Cells [#15407]

Gonzalo Tejera, Martin Llofriu, Alejandra Barrera and Alfredo Weitzenfeld

9:30AM Applying the Canonical Distributed Embodied Evolution Algorithm in a Collective Indoor Navigation Task [#15748]

Pedro Trueba, Abraham Prieto, Francisco Bellas and Duro Richard

9:50AM A Boundedness Theoretical Analysis for GrADP Design: A Case Study on Maze Navigation [#15291]

Zhen Ni, Xiangnan Zhong and Haibo He

10:10AM Neural Network Control for Balancing Performance of a Single-wheel Transportation Vehicle [#15269]

Min S. Ha and Seul Jung

**Session HW1: Hardware 1**
Wednesday, July 15, 9:10AM-10:30AM, Room: Mangerton, Chair: Suri, Manan

9:10AM A Divide-and-Conquer Strategie for FPGA Implementations of Large MLP-based Classifiers [#15392]

Javier Echanobe, Raul Finker and Ines del Campo

9:30AM Noise-Robust Hardware Implementation of Neural Networks [#15497]

Vincent Canals, Miquel L. Alomar, Antoni Morro, Antoni Oliver and Josep L. Rossello

9:50AM Training neural hardware with noisy components [#15702]

Fred Rothganger, Brian Evans, James Aimone and Erik DeBenedictis

10:10AM OXRAM Based ELM Architecture for Multi-Class Classification Applications [#15469]

Manan Suri, Vivek Parmar, Gilbert Sassine and Fabien Alibart

**Panel Session Invited3: Invited Speaker: Cesare Alippi**
Wednesday, July 15, 10:50AM-11:30AM, Room: Auditorium, Chair: Huang, De-Shuang

**Special Session ss12-1: Concept Drift, Domain Adaptation and Learning in Dynamic Environments 1**
Wednesday, July 15, 10:50AM-12:10PM, Room: Ballroom, Chair: Boracchi, Giacomo

10:50AM Credit Card Fraud Detection and Concept-Drift Adaptation with Delayed Supervised Information [#15365]

Andrea Dal Pozzolo, Giacomo Boracchi, Olivier Caelen, Cesare Alippi and Gianluca Bontempi

11:10AM On Sequences of Different Adaptive Mechanisms in Non-Stationary Regression Problems [#15713]

Rashid Bakirov, Bogdan Gabrys and Damien Fay

11:30AM Detecting Anomalous Structures by Convolutional Sparse Models [#15726]

Diego Carrera, Giacomo Boracchi, Alessandro Foi and Brendt Wohlberg

11:50AM Quantifying the Limited and Gradual Concept Drift Assumption [#15826]

Joseph Sarnelle, Anthony Sanchez, Robert Capo, Joshua Haas and Robi Polikar

**Session Spiking2: Spiking neural networks 2**
Wednesday, July 15, 10:50AM-12:10PM, Room: Brehon, Chair: Marsland, John

10:50AM Multi-DL-ReSuMe: Multiple neurons Delay Learning Remote Supervised Method [#15665]

Aboozar Taherkhani, Ammar Belatreche, Yuhua Li and Liam P. Maguire

11:10AM Supervised learning in Spiking Neural Networks with Limited Precision: SNN/LP [#15651]

Evangelos Stromatias and John Marsland

11:30AM Fast-Classifying, High-Accuracy Spiking Deep Networks Through Weight and Threshold Balancing [#15603]

Peter U. Diehl, Daniel Neil, Jonathan Binas, Matthew Cook, Shih-Chii Liu and Michael Pfeiffer

11:50AM Dynamically Evolving Spiking Neural Network for Pattern Recognition [#15538]

Jinling Wang, Ammar Belatreche, Liam Maguire and Martin McGinnity

**Session Robot2: Robotics 2: Sensory and motor processing**
Wednesday, July 15, 10:50AM-12:10PM, Room: Park Suite, Chair: Zhang, LI

10:50AM Estimating Multimodal Attributes for Unknown Objects [#15743]

Daiki Kimura and Osamu Hasegawa

11:10AM Intelligent Facial Expression Recognition with Adaptive Feature Extraction for a Humanoid Robot [#15057]

Kamlesh Mistry, Li Zhang and John Barnden

11:30AM EMG based elbow joint powered exoskeleton for biceps brachii strength augmentation [#15530]

Vladimir Krasim, Gandhi Vaibhav, Yang Zhijun and Karamanoglu Mehmet

11:50AM Population based Mean of Multiple Computations Networks: A Building Block for Kinematic Models [#15727]

Manuel Baum, Martin Meier and Malte Schilling

**Session HW2: Hardware 2**
Wednesday, July 15, 10:50AM-12:10PM, Room: Mangerton, Chair: Rothganger, Fred

10:50AM Memristor Based Neuromorphic Circuit for Ex-Situ Training of Multi-Layer Neural Network Algorithms [#15761]

Chris Yakopcic, Raqibul Hasan and Tarek Taha

11:10AM Efficient Training Algorithms for Neural Networks Based on Memristive Crossbar Circuits [#15720]

Irina Kataeva, Farnood Merrikh-Bayat, Elham Zamanidoost and Dmitri Strukov

11:30AM Design of analog subthreshold encoded neural network circuit in sub-100nm CMOS [#15567]

Benoit Larras, Cyril Lahuec, Fabrice Seguin and Matthieu Arzel

11:50AM A Fully Integrated Analog Neuron for Dynamic Multi-layer Perceptron Networks [#15256]

Melin Ngwar and Jim Wight

**Session ML1: Machine learning 1**
Wednesday, July 15, 11:30AM-12:10PM, Room: Auditorium, Chair: Hakawa, Hiroomi

11:30AM An Autonomous Competitive Learning Algorithm using Quantum Hamming Neural Networks [#15633]

Mohammed Zidan, Alaa Sagheer and Nasser Metwally

11:50AM Vector Classification by a Winner-Take-All Neural Network with Digital Frequency-Locked Loop [#15436]

Hiroomi Hikawa

**Plenary Talk Plenary5: Plenary session: Steve Furber**
Wednesday, July 15, 1:10PM-2:10PM, Room: Auditorium, Chair: Erdi, Peter

**Session ML2: Machine learning 2**
Wednesday, July 15, 2:20PM-4:00PM, Room: Auditorium, Chair: Anguita, Davide

2:20PM Stationarity of Matrix Relevance LVQ [#15248]

Michael Biehl, Barbara Hammer, Frank-Michael Schleif, Petra Schneider and Thomas Villmann

2:40PM Reduction of Catastrophic Forgetting With Transfer Learning and Ternary Output Codes [#15214]

Steven Gutstein and Ethan Stump

3:00PM Efficient Representation Ranking for Transfer Learning [#15262]

Son Tran and Artur Garcez

3:20PM Function approximation for large markov decision processes using self-organizing neural networks [#15475]

Teck-Hou Teng

3:40PM The on-line Curvilinear Component Analysis (onCCA) for Real-Time Data Reduction [#15049]

Giansalvo Cirrincione, Jeanny Herault and Vincenzo Randazzo

**Special Session ss12-2: Concept Drift, Domain Adaptation and Learning in Dynamic Environments 2**
Wednesday, July 15, 2:20PM-4:00PM, Room: Ballroom, Chair: Alippi, Cesare

2:20PM Use of Ensembles of Fourier Spectra in Capturing Recurrent Concepts in Data Streams [#15438]

Sakthithasan Sripirakas, Russel Pears, Albert Bifet and Bernhard Pfahringer

2:40PM A2D2: A Pre-event Abrupt Drift Detection [#15777]

Tatiana Escovedo, Adriano Koshiyama, Marley Vellasco, Andre Abs da Cruz and Rubens Melo

3:00PM An Approach to Handle Concept Drift in Financial Time Series Based on Extreme Learning Machines and Explicit Drift Detection [#15635]

Rodolfo Carneiro Cavalcante and Adriano Lorena Inacio de Oliveira

3:20PM Repeated Play of the SVM Game as a Means of Adaptive Classification [#15646]

Craig Vineyard, Stephen Verzi, Conrad James, James Aimone and Gregory Heileman

3:40PM Combining Offline and Online Classifiers for Life-long Learning [#15577]

Lydia Fischer, Barbara Hammer and Heiko Wersing

**Special Session ss01-1: Computational Intelligence applied to Vision and Robotics (CIVR) 1**
Wednesday, July 15, 2:20PM-4:00PM, Room: Brehon, Chair: Garcia-Rodriguez, Jose

2:20PM Using GNG on 3D Object Recognition in Noisy RGB-D data [#15110]

Jose Carlos Rangel, Vicente Morell, Miguel Cazorla, Sergio Orts-Escolano and Jose Garcia-Rodriguez

2:40PM Evaluation of Multi Feature Fusion at Score-Level for Appearance-based Person Re-Identification [#15122]

Markus Eisenbach, Alexander Kolarow, Alexander Vorndran, Julia Niebling and Horst-Michael Gross

3:00PM ECAS-II: A Hybrid Algorithm for the Construction of Multidimensional Image Segmenters [#15286]

Blanca Priego, Francisco Bellas and Richard Duro

3:20PM Gesture based Human Multi-Robot Interaction [#15381]

Gerard Canal, Cecilio Angulo and Sergio Escalera

3:40PM ChaLearn Looking at People 2015 new competitions: Age Estimation and Cultural Event Recognition [#15482]

Escalera Sergio, Gonzalez Jordi, Baro Xavier, Pardo Pablo, Fabian Junior, Oliu Marc, Escalante Hugo, Huerta Ivan and Guyon Isabelle

**Session RL1: Reinforcement learning 1**
Wednesday, July 15, 2:20PM-4:00PM, Room: Park Suite, Chair: Anderson, Charles

2:20PM Following Newton Direction in Policy Gradient with Parameter Exploration [#15568]

Giorgio Manganini, Matteo Pirotta, Marcello Restelli and Luca Bascetta

2:40PM Faster Reinforcement Learning After Pretraining Deep Networks to Predict State Dynamics [#15781]

Charles Anderson, Minwoo Lee and Daniel Elliott

3:00PM A Comparative Study between Motivated Learning and Reinforcement Learning [#15640]

James Graham, Janusz Starzyk, Zhen Ni, Haibo He, Teck-Hou Teng and Ah-Hwee Tan

3:20PM Continuous-time on-policy neural reinforcement learning of working memory tasks [#15520]

Davide Zambrano, Roelfsema Pieter and Bohte Sander

3:40PM Online Reinforcement Learning by Bayesian Inference [#15242]

Zhongpu Xia and Dongbin Zhao

**Session HW3: Hardware 3**
Wednesday, July 15, 2:20PM-4:00PM, Room: Mangerton, Chair: Roy, Kaushik

2:20PM An efficient SpiNNaker implementation of the Neural Engineering Framework [#15170]

Andrew Mundy, James Knight, Terrence Stewart and Steve Furber

2:40PM Scalable Energy-Efficient, Low-Latency Implementations of Trained Spiking Deep Belief Networks on SpiNNaker [#15502]

Evangelos Stromatias, Daniel Neil, Francesco Galluppi, Michael Pfeiffer, Shih-Chii Liu and Steve Furber

3:00PM Spin-Transfer Torque Magnetic Neuron for Low Power Neuromorphic Computing [#15022]

Abhronil Sengupta and Kaushik Roy

3:20PM Neuron-like Digital Hardware Architecture for Large-scale Neuromorphic Computing [#15641]

Byungik Ahn

3:40PM A neuromorphic hardware framework based on population coding [#15453]

Chetan Singh Thakur, Tara Julia Hamilton, Runchun Wang, Jonathan Tapson and Andre van Schaik

**Session ML3: Machine learning 3**
Wednesday, July 15, 4:20PM-6:00PM, Room: Innisfallen, Chair: Comminiello, Danilo

4:20PM Link prediction in graph construction for supervised and semi-supervised learning [#15384]

Lilian Berton, Jorge Valverde-Rebaza and Alneu de Andrade Lopes

4:40PM Discriminative Concept Learning Network: Reveal High-level Differential Concepts from Shallow Architecture [#15363]

Qiao Wang, Sylvia Young, Aaron Harwood and Cheng Soon Ong

5:00PM Stochastic Discriminant Analysis [#15477]

Mika Juuti, Francesco Corona and Juha Karhunen

5:20PM Improved Error Bounds Based on Worst Likely Assignments [#15676]

Eric Bax

5:40PM Learning the Hash Code with Generalised Regression Neural Networks for Handwritten Signature Biometric Data Retrieval [#15617]

Bernardete Ribeiro, Noel Lopes and Catarina Silva

**Special Session ss12-3: Concept Drift, Domain Adaptation and Learning in Dynamic Environments 3**
Wednesday, July 15, 4:20PM-6:00PM, Room: Ballroom, Chair: Lemaire, Vincent

4:20PM Monitoring Term Drift Based on Semantic Consistency in an Evolving Vector Field [#15693]

Peter Wittek, Sandor Daranyi, Efstratios Kontopoulos, Theodoros Moysiadis and Ioannis Kompatsiaris

4:40PM Concept Drift Detection using Supervised Bivariate Grids [#15273]

Christophe Salperwyck, Marc Boulle and Vincent Lemaire

5:00PM Comparison Between Inverse Modelling and Data Assimilation to Estimate Rainfall from Runoff Using the Multilayer Perceptron [#15228]

Anne Johannet, Virgile Taver, Valerie Borrell Estupina, Marc Vinches, Severin Pistre and Dominique Bertin

5:20PM Feature Ranking in Changing Environments where New Features are Introduced [#15373]

Alexandra Degeest, Michel Verleysen and Benoit Frenay

5:40PM Concept Drift Detection for Streaming Data [#15186]

Zubin Abraham and Heng Wang

**Special Session ss01-2: Computational Intelligence applied to Vision and Robotics (CIVR) 2**
Wednesday, July 15, 4:20PM-6:00PM, Room: Brehon, Chair: Azorin-Lopez, Jorge

4:20PM Processing Point Cloud Sequences with Growing Neural Gas [#15620]

Sergio Orts-Escolano, Jose Garcia-Rodriguez, Vicente Morell, Miguel Cazorla, Marcelo Saval-Calvo and Jorge Azorin

4:40PM Non-rigid point set registration using color and data downsampling [#15692]

Marcelo Saval-Calvo, Sergio Orts-Escolano, Jorge Azorin-Lopez, Jose Garcia-Rodriguez, Andres Fuster-Guillo, Vicente Morell-Gimenez and Miguel Cazorla

5:00PM Self-Organizing Activity Description Map to Represent and Classify Human Behaviour [#15718]

Jorge Azorin-Lopez, Marcelo Saval-Calvo, Andres Fuster-Guillo, Jose Garcia-Rodriguez and Sergio Orts-Escolano

5:20PM Enhanced Image Classification With a Fast-Learning Shallow Convolutional Neural Network [#15736]

Mark McDonnell and Tony Vladusich

5:40PM Improving Bag of Visual Words Representations with Genetic Programming [#15739]

Hugo Jair Escalante, Jose Martinez, Sergio Escalera, Victor Ponce and Xavier Baro

**Session RL2: Reinforcement learning 2**
Wednesday, July 15, 4:20PM-6:00PM, Room: Park Suite, Chair: Potter, Steve

4:20PM Learning Eye Movements Strategies on Tiled Large High-Resolution Displays using Inverse Reinforcement Learning [#15571]

Redwan Abdo A. Mohammed and Oliver Staadt

4:40PM Interactive Reinforcement Learning through Speech Guidance in a Domestic Scenario [#15293]

Francisco Cruz, Johannes Twiefel, Sven Magg, Cornelius Weber and Stefan Wermter

5:00PM Approximate Policy Iteration with Unsupervised Feature Learning based on Manifold Regularization [#15034]

Hongliang Li, Derong Liu and Ding Wang

5:20PM A Spiking Neuronal Model Learning a Motor Control Task by Reinforcement Learning and Structural Synaptic Plasticity [#15358]

Spueler Martin, Nagel Sebastian and Rosenstiel Wolfgang

5:40PM Adaptive-Critic-Based Control of a Synchronous Generator in a Power System Using Biologically Inspired Artificial Neural Networks [#15634]

Jing Dai, Ganesh K. Venayagamoorthy, Ronald G. Harley, Yi Deng and Steve M. Potter

**Session Noise: Anomaly and noise**
Wednesday, July 15, 4:20PM-6:00PM, Room: Mangerton, Chair: Angelov, Plamen

4:20PM Expected Similarity Estimation for Large Scale Anomaly Detection [#15066]

Markus Schneider, Wolfgang Ertel and Guenther Palm

4:40PM Self-structured Confabulation Network for Fast Anomaly Detection and Reasoning [#15135]

Qiuwen Chen, Qing Wu, Morgan Bishop, Richard Linderman and Qinru Qiu

5:00PM The Adaptable Buffer Algorithm for High Quantile Estimation in Non-Stationary Data Streams [#15041]

Ognjen Arandjelovic, Pham Duc-Son and Venkatesh Svetha

5:20PM PCAGA: Principal Component Analysis Based Genetic Algorithm for Solving Conditional Nonlinear Optimal Perturbation [#15360]

Bin Mu, Linlin Zhang, Shijin Yuan and Hongyu Li

5:40PM Online Fault Detection Based on Typicality and Eccentricity Data Analytics [#15626]

Bruno Costa, Clauber Bezerra, Luiz Affonso Guedes and Plamen Angelov

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## Thursday, July 16, 2015

**Plenary Talk Plenary6: Plenary session: Lee Giles**
Thursday, July 16, 8:00AM-9:00AM, Room: Auditorium, Chair: Minai, Ali

**Special Session ss16-1: Computational Intelligence Algorithms for Digital Audio Applications 1**
Thursday, July 16, 9:10AM-10:30AM, Room: Auditorium, Chair: Squartini, Stefano

9:10AM Non-Linear Prediction with LSTM Recurrent Neural Networks for Acoustic Novelty Detection [#15683]

Erik Marchi, Fabio Vesperini, Felix Weninger, Florian Eyben, Stefano Squartini and Bjoern Schuller

9:30AM Polyphonic Sound Event Detection Using Multi Label Deep Neural Networks [#15500]

Emre Cakir, Toni Heittola, Heikki Huttunen and Tuomas Virtanen

9:50AM An Interactive Optimization Procedure for Stereophonic Acoustic Echo Cancellation Systems [#15252]

Laura Romoli, Stefania Cecchi, Francesco Piazza, Danilo Comminiello, Michele Scarpiniti and Aurelio Uncini

10:10AM Functional Link Expansions for Nonlinear Modeling of Audio and Speech Signals [#15251]

Danilo Comminiello, Simone Scardapane, Michele Scarpiniti, Raffaele Parisi and Aurelio Uncini

**Session ML4: Machine learning 4**
Thursday, July 16, 9:10AM-10:30AM, Room: Ballroom, Chair: Bertini, Joao

9:10AM Refining Constructive Neural Networks Using Functionally Expanded Input Data [#15299]

Joao Bertini and Maria Nicoletti

9:30AM Exponential C-loss for Data Fitting [#15618]

Badong Chen, Ren Wang, Nanning Zheng and Jose Principe

9:50AM Kolmogorov Complexity Vector: A Novel Data Representation [#15394]

Ge Yang and Ali Ghodsi

10:10AM A Sample Partition Method for Learning to Rank Based on Query-level Vector Extraction [#15368]

Jungang Xu, Shilong Zhou, Hong Chen and Pengfei Li

**Session Spiking3: Spiking neural networks 3**
Thursday, July 16, 9:10AM-10:30AM, Room: Brehon, Chair: TBA

9:10AM Runtime Detection of Activated Polychronous Neuronal Group towards its Spatiotemporal Analysis [#15207]

Haoqi Sun, Yan Yang, Olga Sourina and Guang-Bin Huang

9:30AM Scale and Translation Invariant Learning of Spatio-Temporal Patterns using Longest Common Subsequences and Spiking Neural Networks [#15737]

Banafsheh Rekabdar, Monica Nicolescu, Mircea Nicolescu and Richard Kelley

9:50AM GPU-based Fast Parameter Optimization for Phenomenological Spiking Neural Models [#15561]

Zafeirios Fountas and Murray Shanahan

10:10AM Stochastic and Asynchronous Spiking Dynamic Neural Fields [#15709]

Benoit Chappet de Vangel, Cesar Torres-Huitzil and Bernard Girau

**Session Cluster: Clustering**
Thursday, July 16, 9:10AM-10:30AM, Room: Park Suite, Chair: Wunsch, Donald C.

9:10AM The Unbalancing Effect of Hubs on K-medoids Clustering in High-Dimensional Spaces [#15019]

Dominik Schnitzer and Arthur Flexer

9:30AM Spectral Clustering Using Robust Similarity Measure Based on Closeness of Shared Nearest Neighbors [#15316]

Xiucai Ye and Tetsuya Sakurai

9:50AM Multi-Prototype Local Density-based Hierarchical Clustering [#15747]

Leonardo Enzo Brito da Silva and Donald C. Wunsch II

10:10AM Impact of different Metrics on Multi-View Clustering [#15253]

Angela Serra, Dario Greco and Roberto Tagliaferri

**Session BioPerc1: Biologically inspired perception 1**
Thursday, July 16, 9:10AM-10:30AM, Room: Mangerton, Chair: Khan Iftekharuddin

9:10AM Modelling Retinal Ganglion Cells using Self-Organising Fuzzy Neural Networks [#15604]

Scott McDonald, Dermot Kerr, Sonya Coleman, Martin McGinnity and Philip Vance

9:30AM HEVS: A Hierarchical Computational Model for Early Stages of the Visual System [#15126]

Jiuqi Han, Qingqun Kong, Yi Zeng and Hongwei Hao

9:50AM Design of a Silicon Cochlea System with Biologically Faithful Response [#15787]

Shiwei Wang, Thomas Koickal, Godwin Enemali, Luiz Gouveia, Lei Wang and Alister Hamilton

10:10AM A Bio-Inspired Method for Object Representation [#15141]

Hui Wei

**Special Session ss16-2/r: Computational Intelligence Algorithms for Digital Audio Applications 2 + Regular session on Image Analysis**
Thursday, July 16, 10:50AM-12:10PM, Room: Auditorium, Chair: Squartini, Stefano; Mello, Carlos

10:50AM A Deep Neural Network Approach for Voice Activity Detection in Multi-Room Domestic Scenarios [#15340]

Giacomo Ferroni, Roberto Bonfigli, Emanuele Principi, Stefano Squartini and Francesco Piazza

11:10AM Data-driven vocal folds models for the representation of both acoustic and high speed video data [#15586]

Carlo Drioli and Gian Luca Foresti

11:30AM Text Segmentation in Ancient Topographic Maps and Floor Plans with Support Vector Data Description [#15331]

Saulo Machado and Carlos Mello

11:50AM One-shot Training of Polynomial Cellular Neural Networks and Applications in Image Processing. [#15133]

Antonio Arista-Jalife and Eduardo Gomez-Ramirez

**Session Datamine: Data and text mining**
Thursday, July 16, 10:50AM-12:10PM, Room: Ballroom, Chair: Schikuta, Erich

10:50AM Kernel Spectral Document Clustering Using Unsupervised Precision-Recall Metrics [#15543]

Raghvendra Mall and Johan Suykens

11:10AM Using Active Learning Techniques for Improving Database Schema Matching Methods [#15509]

Diego Rodrigues, Altigran Silva, Rosiane Rodrigues and Eulanda Santos

11:30AM Directed Generalized Measure of Association: A Data Driven Approach Towards Causal Inference [#15762]

Mehrnaz Khodam Hazrati, Andreas Keil and Jose C. Principe

11:50AM Semantic Extensions to the Vienna Neural Network Specification Language [#15560]

Erich Schikuta, Altaf Huqqani and Thomas Kopica

**Session Image: Image analysis**
Thursday, July 16, 10:50AM-12:10PM, Room: Brehon, Chair: Lin, Feng

10:50AM Multi-Scale Local Shape Analysis and Feature Selection in Machine Learning Applications [#15229]

Paul Bendich, Ellen Gasparovic, John Harer, Rauf Izmailov and Linda Ness

11:10AM Pixel Characteristics based Feature Extraction Approach for Roadside Object Detection [#15463]

Sujan Chowdhury, Brijesh Verma, Mary Tom and Mengjie Zhang

11:30AM Dual Spatial Pyramid On Rotation Invariant Texture Feature For HEp-2 Cell Classification [#15137]

Xiang Xu, Feng Lin, Carol Ng and Khai pang Leong

11:50AM Factor Graphs for Pixelwise Illuminant Estimation [#15759]

Lawrence Mutimbu and Antonio Robles-Kelly

**Session Local : Local learning**
Thursday, July 16, 10:50AM-12:10PM, Room: Park Suite, Chair: Zhang, Haijun

10:50AM Self-Organizing Map-based Probabilistic Associative Memory for Sequential Patterns [#15527]

Niitsuma Jun and Osana Yuko

11:10AM Learning of local predictable representations in partially learnable environments [#15680]

Mathieu Lefort and Alexander Gepperth

11:30AM A Tree-Structured Representation for Book Author and Its Recommendation Using Multilayer SOM [#15369]

Lu Lu and Haijun Zhang

11:50AM An ART-like Algorithm for constructing RBF Neural Networks [#15305]

Xi Meng, Junfei Qiao and Honggui Han

**Session BioPerc2: Biologically inspired perception 2**
Thursday, July 16, 10:50AM-12:10PM, Room: Mangerton, Chair: Smith, Leslie

10:50AM C.elegans chemotaxis inspired neuromorphic circuit for contour tracking and obstacle avoidance [#15533]

Shibani Santurkar and Bipin Rajendran

11:10AM Gabor feature processing in spiking neural networks from retina-inspired data [#15108]

Aristeidis Tsitiridis, Cristina Conde, Isaac Martin de Diego, Jose Sanchez del Rio Saez, Jorge Raul Gomez and Enrique Cabello

11:30AM Novel Hierarchical Cellular Simultaneous Recurrent Neural Network for Object Detection [#15297]

Mahbubul Alam, Lasitha Vidyaratne and Khan Iftekharuddin

11:50AM A Biologically Inspired Onset and Offset Sound Segmentation Approach [#15097]

Andrew Abel, Dean Hunter and Leslie Smith

**Workshop ws1: Workshop 1: The 2nd International Workshop on Advances in Learning from/with Multiple Learners (ALML 2015)**
Thursday, July 16, 1:30PM-4:30PM, Room: Ballroom, Chair: Nistor Grozavu and Guenael Cabanes

1:30PM Uncorrelated Transferable Feature Extraction for Signal Classification in Brain-Computer Interfaces [#15819]

Honglei Shi, Jinhua Xu and Shiliang Sun

**Workshop ws2: Workshop 2: The International Workshop on Spatial Representations in Biology and Robots**
Thursday, July 16, 1:30PM-4:30PM, Room: Brehon, Chair: Jorg Conradt and Fred Hamker

**Workshop ws3: Workshop 3: Computational Neurology and Psychiatry: Do we need it?**
Thursday, July 16, 1:30PM-4:30PM, Room: Park Suite, Chair: Basabdatta Sen Bhattacharya and Peter Erdi

**Workshop ws4: Workshop 4: BMI Workshop on Brain-Mind 1**
Thursday, July 16, 1:30PM-4:30PM, Room: Mangerton, Chair: Leonid Perlovsky and Juyang (John) Weng

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## Friday, July 17, 2015

**Workshop ws5-1: Workshop 5: The 2nd International Workshop on Computational Energy Management in Smart Grids (CEMiSG 2015) 1**
Friday, July 17, 9:00AM-12:00PM, Room: Innisfallen, Chair: Stefano Squartini

9:00AM A Learning Intelligent System for Fault Detection in Smart Grid by a One-Class Classification Approach [#15682]

Enrico De Santis, Antonello Rizzi, Alireza Sadeghian and Frattale Mascioli Fabio Massimo

9:20AM Energy management with the support of dynamic pricing strategies in real micro-grid scenarios [#15493]

Marco Severini, Stefano Squartini, Marco Fagiani and Francesco Piazza

9:40AM Appliance Level Demand Modeling and Pricing Optimization for Demand Response Management in Smart Grid [#15233]

Fan-Lin Meng and Xiao-Jun Zeng

10:00AM Unit Commitment Considering Multiple Charging and Discharging Scenarios of Plug-in Electric Vehicles [#15254]

Zhile Yang, Kang Li, Qun Niu and Aoife Foley

10:20AM Methods for Clustering the Eletrical Load in European Countries [#15064]

Ankit Kumar Tanwar, Emanuele Crisostomi, Pietro Ferraro, Giuseppe Giunta, Marco Raugi and Mauro Tucci

**Workshop ws5-2: Workshop 5: The 2nd International Workshop on Computational Energy Management in Smart Grids (CEMiSG 2015) 2**
Friday, July 17, 1:30PM-4:30PM, Room: Innisfallen , Chair: Stefano Squartini

1:30PM Thermal Comfort Control Based on MEC Algorithm for HVAC Systems [#15240]

Dong Li, Dongbin Zhao, Yuanheng Zhu and Zhongpu Xia

1:50PM Indoor thermal comfort control through fuzzy logic PMV optimization [#15607]

Lucio Ciabattoni, Gionata Cimini, Francesco Ferracuti, Massimo Grisostomi, Gianluca Ippoliti and Matteo Pirro

2:10PM A Non-Linear State Space Frequency Estimator for Three-Phase Power Systems [#15591]

Sayed Pouria Talebi, Sithan Kanna and Danilo Mandic

2:30PM A Novelty Detection approach to identify the occurrence of leakage in Smart Gas and Water Grids [#15289]

Marco Fagiani, Stefano Squartini, Marco Severini and Francesco Piazza

**Workshop ws-canceled: Canceled Workshop: IJCNN 2015 Workshop on Automatic Machine Learning - AutoML**
Friday, July 17, 1:30PM-4:30PM, Room:

1:30PM Design of the 2015 ChaLearn AutoML Challenge [#15695]

Isabelle Guyon, Kristin Bennett, Gavin Cawley, Hugo Jair Escalante, Sergio Escalera, Tin Kam Ho, Nuria Macia, Bisakha Ray, Alexander Statnikov, Evelyne Viegas and Merhreen Saeed

# 10 Author index

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- The index only includes authors of papers that appear in the proceedings.

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# 11 Errata

Errors that have been identified in the PDF papers are indicated below.

| Paper ID: | Location | Original | Corrections/Additions |
|---|---|---|---|
| 15335 | Fig. 9 (a)–(c), $x$-axis label | b | $\beta$ |
| 15363 | Acknowledgments | | This work has received the Google Ph.D. travel prize from Google Australia. |
| 15384 | Equation 2 | $s^c_{v_i,v_j} = \|\Gamma(v_i)\| \cap \|\Gamma(v_j)\|$ | $s^c_{v_i,v_j} = \|\Gamma(v_i) \cap \Gamma(v_j)\|$ |
| 15833 | Acknowledgments | National Science Foundation under grant #19200000000036 | National Science Foundation under grant #1201790 |
| 15280 | Pg 5 (right col) Line 26 | on ten datasets | on nine datasets |
| | Pg 5 (right col)Line 27 | on two datasets | on three datasets |
| | Pg 5 (right col) Line 29 | on 11 datasets | on 10 datasets |
| | Pg 5 (right col) Line 30 | on the ALL dataset | on the ALL and BrainTumor datasets |
| | Pg 5 (right col) Line 32 | on 10 datasets, and NMFK-means and SRSC obtain the best RI on DLBCL dataset and ALL dataset, respectively. | on 9 datasets, and SRSC obtains the best RI on ALL and BrainTumor datasets. |
| | Pg 5 (right col) Line 36 | 2.10% | 1.47% |
| | Pg 5 (right col) Line 37 | 4.26%, 3.25% and 2.42% | 3.73%, 2.54% and 2.22% |
| | Pg 5 (right col) Line 40 | 4.40% | 4.62% |
| | Pg 6 Tab. II row 11 col 6 | 0.4324 | 0.5081 |
| | Pg 6 Tab. II row 14 col 6 | 0.7837 | 0.7900 |
| | Pg 6 Tab. III row 11 col 6 | 0.2117 | 0.2750 |
| | Pg 6 Tab. III row 14 col 6 | 0.5119 | 0.5172 |
| | Pg 6 Tab. IV row 11 col 6 | 0.3951 | 0.4804 |
| | Pg 6 Tab. IV row 14 col 6 | 0.7743 | 0.7814 |
| | Pg 7 Tab. V row 11 col 6 | 0.6477 | 0.6713 |
| | Pg 7 Tab. V row 14 col 6 | 0.7772 | 0.7792 |
| | Pg 7 Tab. VI row 11 col 6 | 0.5802 | 0.6138 |
| | Pg 7 Tab. VI row 14 col 6 | 0.4140 | 0.4168 |
| | Pg 7 (left col) line 6 | Besides the above four metrics | Besides the above five metrics |

# 12 Venue Floor Plan

## 12.1 Killarney Convention Center Complex Overview

Brehon

Park Suite

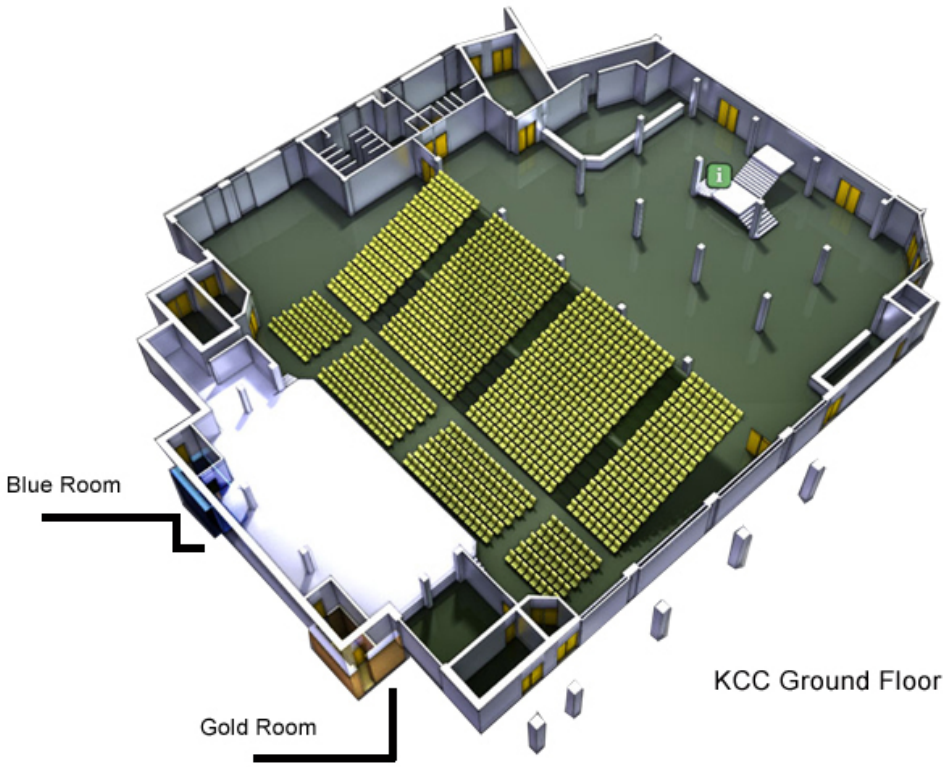
INEC Auditorium

Ballroom

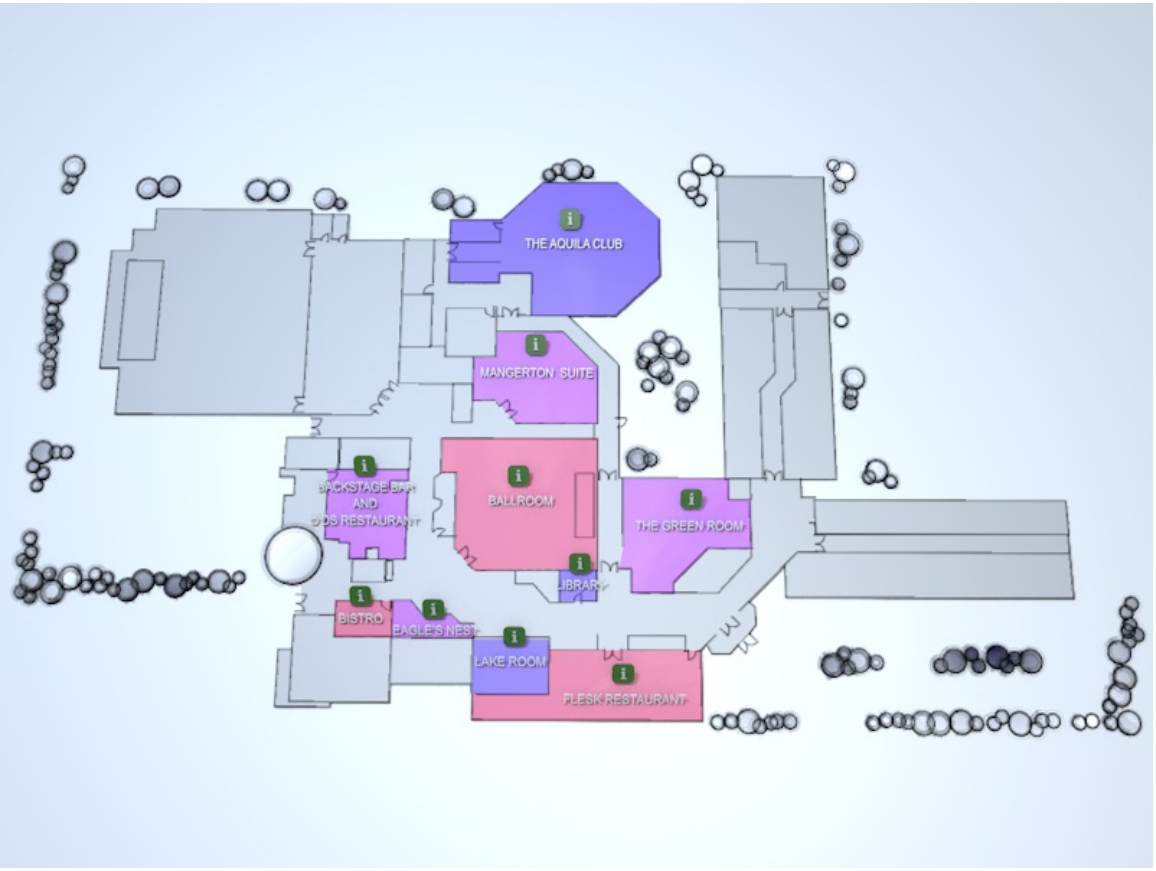
Mangerton

Torc/Ross/Innisfallen

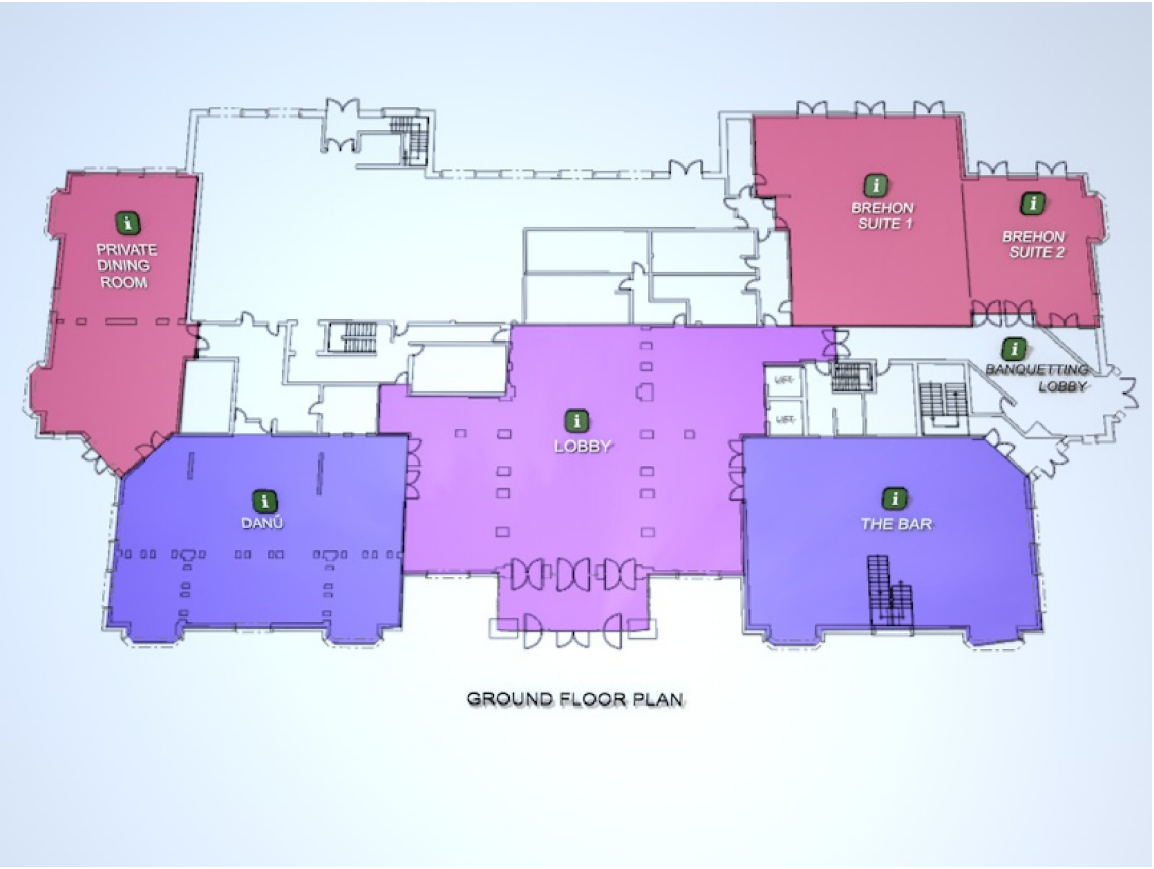
## 12.2 Auditorium: Convention Center Ground Floor



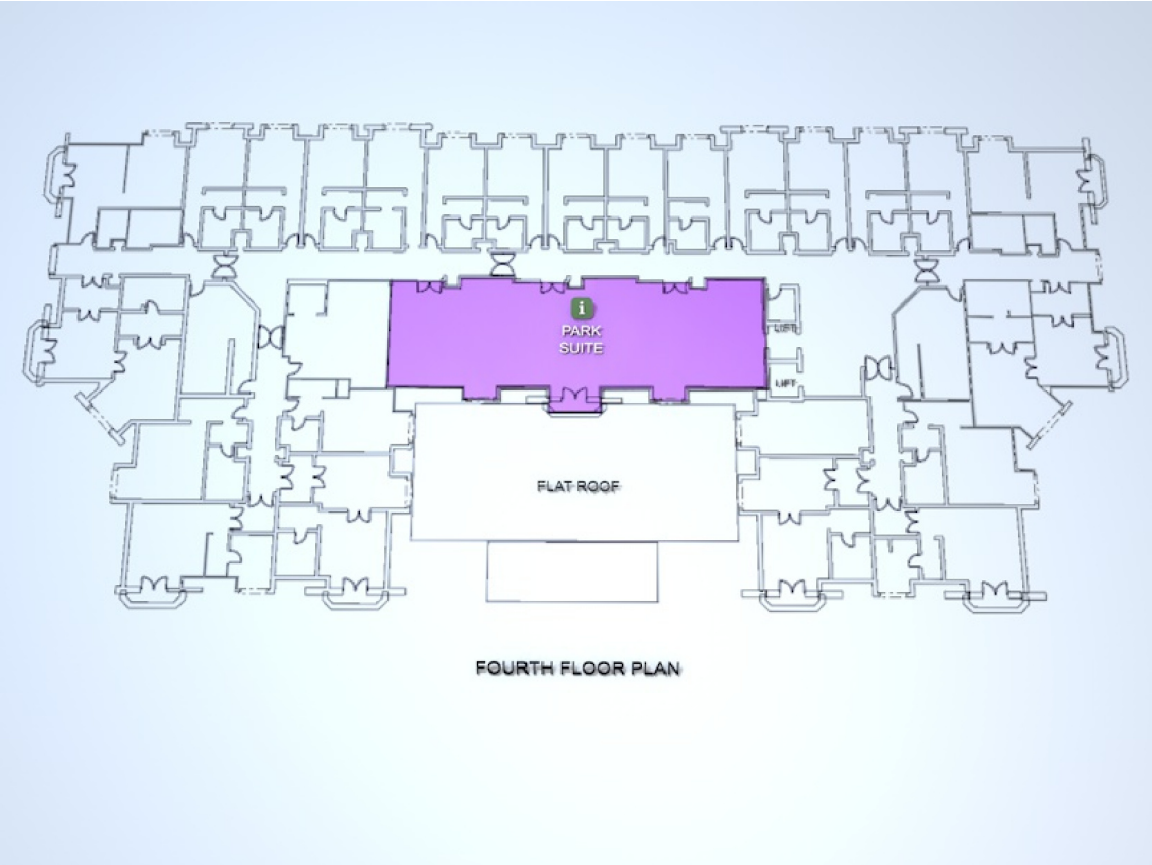
## 12.3 Ballroom: Glenneagle Ground Floor



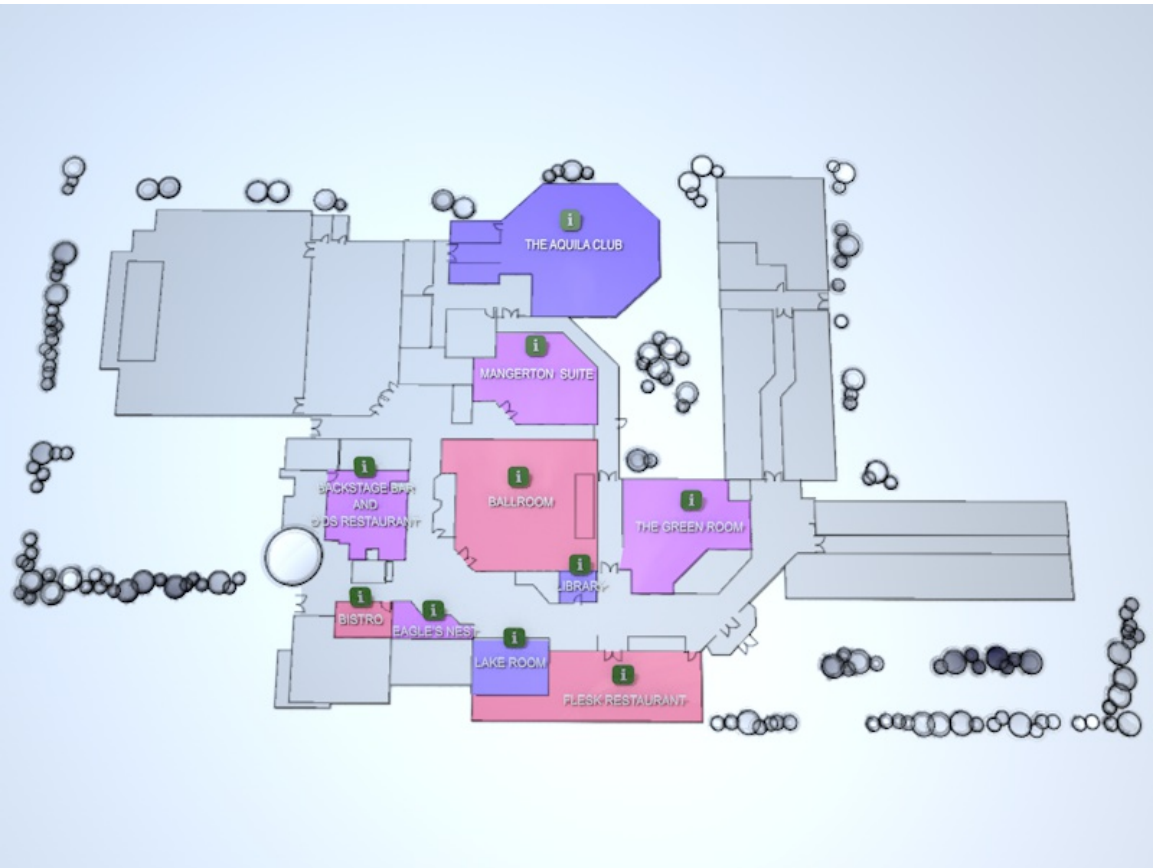

## 12.4 Brehon: Brehon Hotel Ground Floor



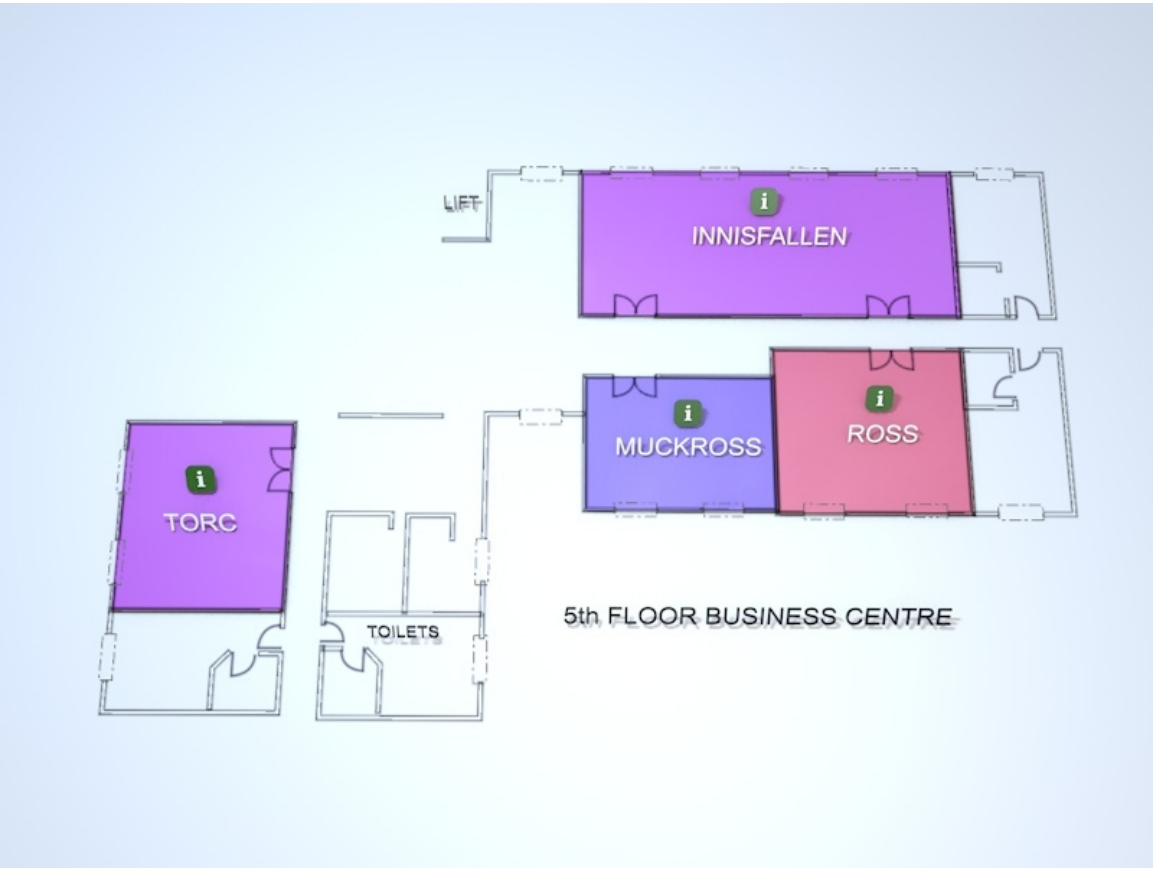
## 12.5 Park Suite: Brehon Hotel 4th Floor




## 12.6 Mangerton: Gleneagle Ground Floor



## 12.7 Torc, Ross, Innisfallen: Gleneagle 5th floor



# IEEE World Congress on Computational Intelligence

## 25-29 July 2016, Vancouver, Canada

IEEE WCCI 2016
Vancouver Canada

The IEEE World Congress on Computational Intelligence (IEEE WCCI) is the largest technical event in the field of computational intelligence. The IEEE WCCI 2016 will host three conferences: The 2016 International Joint Conference on Neural Networks (IJCNN 2016), the 2016 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE 2016), and the 2016 IEEE Congress on Evolutionary Computation (IEEE CEC 2016) under one roof. It encourages cross-fertilization of ideas among the three big areas and provides a forum for intellectuals from all over the world to discuss and present their research findings on computational intelligence.

IEEE WCCI 2016 will be held at the Vancouver Convention Centre, Vancouver, Canada. Vancouver is Canada's Pacific gem, offering a winning combination of world-class hotels, meeting venues, and restaurants in a setting of spectacular beauty. Few convention cities can offer such a wide range of cosmopolitan amenities in a downtown core that is safe, clean, pedestrian friendly, and stunning in its backdrop of mountains and ocean.

IJCNN is the flagship conference of the International Neural Network Society and the IEEE Computational Intelligence Society. It covers a wide range of topics in the field of neural networks, from biological neural network modeling to artificial neural computation.

FUZZ-IEEE is the foremost conference in the field of fuzzy systems. It covers all topics in fuzzy systems, from theory to applications.

IEEE CEC is a major event in the field of evolutionary computation, and covers all topics in evolutionary computation from theory to applications.

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Papers for IEEE WCCI 2016 should be submitted electronically through the Congress website at www.wcci2016.org, and will be refereed by experts in the fields and ranked based on the criteria of originality, significance, quality and clarity.

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IEEE WCCI 2016 will feature pre-Congress tutorials, covering fundamental and advanced topics in computational intelligence. A tutorial proposal should include title, outline, expected enrollment, and presenter/organizer biography. Inquiries regarding tutorials should be addressed to Tutorials Chairs.

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IEEE WCCI 2016 solicits proposals for special sessions within the technical scope of the three conferences. Special sessions, to be organized by internationally recognized experts, aim to bring together researchers in special focused topics. Cross-fertilization of the three technical disciplines and newly emerging research areas are strongly encouraged. Inquiries regarding special sessions and proposals should be addressed to Special Sessions Chairs.

### Call for Competition Proposals

IEEE WCCI 2016 will host competitions to stimulate research in computational intelligence. A competition proposal should include descriptions of the problem(s) addressed, evaluation procedures, and a biography of the organizers. Inquiries regarding competitions should be addressed to the Competitions Chair.

### Important Dates

- 15 Nov 2015 — Special Session & Workshop Proposals Deadline
- 15 Dec 2015 — Competition & Tutorial Proposals Deadline
- 15 Jan 2016 — Paper Submission Deadline
- 15 Mar 2016 — Paper Acceptance Notification Date
- 15 Apr 2016 — Final Paper Submission & Early Registration Deadline
- 25-29 Jul 2016 — IEEE WCCI 2016 Vancouver, Canada

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